

Discount factors and monetary policy: Evidence from dual-listed stocks[☆]Quentin Vandeweyer^{a,*,}, Minghao Yang^b, Constantine Yannelis^c^a University of Chicago, Booth School of Business, 5807 S Woodlawn Ave, Chicago, 60637, IL, USA^b University of California Berkeley, Haas School of Business, 2220 Piedmont Ave, Berkeley, 94720, CA, USA^c University of Cambridge, Austin Robinson Building, Sidgwick Ave, Cambridge, CB3 9DD, United Kingdom

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ABSTRACT

This paper studies the transmission of monetary policy to the stock market through investors' discount factors. To isolate this channel, we investigate the effect of US monetary policy surprises on the ratio of prices of the same stock listed simultaneously in Hong Kong and Mainland China. We identify a strong discount rate channel driven exclusively by cycle-amplifying surprises, defined as rate cuts during easing cycles and surprise hikes during tightening cycles. A 100 basis point of such cycle-amplifying surprise induces a 30 basis point change in the price ratio within five days.

1. Introduction

The effect of monetary policy on stock prices is a critical component of the transmission mechanism to the economy and remains a central question in finance. This channel can be decomposed into two key components. First, monetary policy can influence asset prices by shaping expectations about firms' future cash flows, either by directly boosting profits through increased aggregate demand (Caballero and Simsek, 2022, 2023) or by releasing new information about the economy (Nakamura and Steinsson, 2018; Gómez-Cram and Grotteria, 2022). Second, monetary policy can alter how investors discount these cash flows over time and across states of the world. For instance, an interest rate cut may lead investors to lower the risk premia they demand for holding risky stocks, either because they are more willing to take on risk to meet return targets (i.e., "reaching for yield", as in Hanson and Stein (2015)) or because they revise upward their expectations about the central bank's responsiveness to macroeconomic and financial conditions (Cieslak and Pang, 2021). Alternatively, monetary policy can affect investors' time discount rates (Nagel and Xu, 2024). Understanding the behavior of discount factors is crucial for evaluating the plausibility of macro-financial models (Cochrane, 2017), assessing the impact of monetary policy on financial intermediation (Begenau et al., 2015), and analyzing its macroeconomic transmission through the stock market (Kekre and Lenel, 2022). However, studying discount rate shocks is challenging because they are often confounded with news about firms' cash flows.

This paper provides new insights into how monetary policy impacts stock prices through investors' discount factors, using dual-listed stocks in Mainland China and Hong Kong that represent claims to the same cash flows in two segmented markets. First, we offer conclusive evidence that monetary policy influences stock prices beyond its direct effect on firms' cash flows, operating through revisions in investors' discount factors. Second, we show that these revisions are asymmetric, driven primarily by *amplifying* surprises—surprise rate cuts during easing cycles and surprise rate hikes during tightening cycles—while *contradictory* surprises—such as rate cuts during tightening cycles or rate hikes during easing cycles—produce insignificant effects. Third, we find that high-frequency strategies may underestimate the effects of monetary policy, likely due to the simultaneous release of information about cash flows. These findings suggest that model calibrations and policy rules may need to be updated to account for the stronger effects of monetary policy on asset prices compared to those commonly reported in the literature.

In this study, we propose a new approach to isolate and study the discount rate channel by exploiting a unique institutional feature of the Chinese stock market, where some companies' stocks are listed simultaneously in Mainland China and Hong Kong. The existence of capital controls, which prevent flows between the two markets, implies that the stock prices of the same company can diverge across exchanges despite identical underlying fundamentals (Carpenter et al., 2020; Jia et al., 2017; Mei et al., 2009). In contrast, geographic and cultural

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* Corresponding author.

E-mail addresses: quentin.vandeweyer@chicagobooth.edu (Q. Vandeweyer), mhyang@haas.berkeley.edu (M. Yang), cny21@cam.ac.uk (C. Yannelis).

proximity allows for the free exchange of information, including news about firms' future cash flows. Leveraging this duality, we propose a difference-in-differences empirical design to isolate the discount rate channel of monetary policy on stock prices. Specifically, we examine the effect of US monetary policy surprises, as introduced by Kuttner (2001) and Bernanke and Kuttner (2005), on the ratio of the prices of the same stock listed in both Mainland China and Hong Kong—the A/H ratio.

This design offers three main advantages. First, by focusing on the ratio of the prices of the same company across the two exchanges, we neutralize the effect of revisions in cash flow expectations, which should impact both A and H shares homogeneously under a common information set. This specification thus allows us to isolate the effect of monetary policy on stocks that arises from revisions in discount factors. Second, unlike conventional event studies that focus on the US stock market—where US monetary policy may react to new macroeconomic information (Bauer and Swanson, 2023) or directly to stock market dynamics (Rigobon and Sack, 2003; Cieslak and Vissing-Jorgensen, 2021)—US monetary policy is plausibly exogenous to the economic conditions in Hong Kong. Nonetheless, because the Hong Kong dollar (HKD) is pegged to the US dollar (USD), the Hong Kong overnight interbank rate (HIBOR) closely tracks the federal funds rate and is highly sensitive to US monetary policy surprises. These two features create an ideal setting to study the effect of monetary policy on stock prices without the endogeneity issues inherent in studies of US monetary policy and the US stock market. Third, by examining the ratio of stock prices based on the same fundamentals, our difference-in-differences design reduces noise by filtering out fundamental news about firms' cash flows, thereby increasing the precision of our estimates. This approach enables us to identify significant effects of monetary policy surprises on discount rates, even beyond Federal Open Market Committee (FOMC) announcement days.

The main contribution of our analysis is to demonstrate the existence of a substantial *discount factor* transmission channel of monetary policy to the stock market¹ and provide evidence against the view that central banks do not affect asset prices.² Our estimates indicate that a 25-basis-point surprise decrease in the US federal funds rate leads to a 3% increase in the price of Chinese stocks listed in Hong Kong relative to the same company's stock listed in Mainland China over the subsequent five days. This effect is notable when compared to the 1% figure reported by Bernanke and Kuttner (2005) for the overall impact of monetary policy surprises on stock prices.

Moreover, we find that the isolated discount factor effect of monetary policy on stock prices is asymmetric, with only surprises that amplify the concurrent cycle – such as surprise rate cuts during an easing cycle or surprise hikes during a tightening cycle – resulting in statistically significant changes in the A/H ratio. These patterns align with models in which such shocks provide particularly strong signals about the Federal Reserve's (the Fed) propensity to respond to adverse market and macroeconomic conditions. Such a mechanism is consistent with Bauer et al. (2022) providing evidence from panel data on professional forecasts that public perceptions of the relationship between the federal funds rate and economic conditions are highly time-varying. Cieslak et al. (2019) further document substantial reductions in the equity risk premium following unexpectedly accommodative

surprises during the 1994–2016 period. If investors interpret surprise rate cuts as evidence of the Fed reacting more aggressively to negative shocks, they lower the risk premia required to hold stocks, leading to an increase in stock prices—a phenomenon often referred to as the “Fed Put” effect. In the same 1994–2016 sample, Cieslak and Vissing-Jorgensen (2021) provide textual analysis evidence suggesting that the Fed is indeed attentive and reactive to stock market conditions, while Cieslak and Pang (2021) show that reductions in risk premia are 3.5 times larger than orthogonalized surprises to short-term rates in a VAR framework. Extending the sample to include two additional tightening cycles (2016–2019 and 2022–2024), our findings suggest that while investors have positively updated their expectations about the Fed's willingness to respond to adverse news during easing cycles, they have also negatively adjusted these expectations when the Fed hiked beyond market expectations during recent tightening cycles. Hadad et al. (2025) provide complementary evidence suggesting that a substantial portion of central bank asset purchase effects on asset prices arises from investors positively revising their expectations regarding future central bank support in adverse economic conditions.

We further investigate how monetary policy surprises influence the discount factor across four stock characteristics. First, following Fama and French (1992), we examine how a stock's market risk (measured by its Capital Asset Pricing Model (CAPM) beta), size (measured by market capitalization), and value (measured by the ratio of market capitalization to earnings) shape its response to monetary policy surprises. We find that high-beta stocks are significantly more reactive to these surprises. This result aligns with the theory that monetary policy surprises impact prices through changes in risk premia as, according to standard asset pricing models, a revision in risk premia has a disproportionately stronger effect on the prices of riskier stocks.

We also find evidence that small firms—characterized by low market capitalization—and value firms—characterized by low price-to-earnings ratios—are more sensitive to discount factor adjustments following FOMC announcements. This finding is consistent with the hypothesis that such firms require a larger macroeconomic announcement risk premium, as proposed by the distressed-firm explanation of the value premium (Fama and French, 1992; Chan and Chen, 1991), and aligns with previous research showing that discount factor fluctuations primarily affect near-term cash flows rather than those further in the future (Binsbergen et al., 2012; Binsbergen and Koijen, 2017). Finally, we interact the monetary policy surprise with the lagged A/H ratio for each stock and find that stocks with the highest A/H ratios are the most affected by monetary policy surprises.

To conclude, we provide a series of additional tests to assess the robustness of those results. We perform two additional exercises to ensure that heterogeneous revisions in firms' cash flow expectations are not driving our results—as could be possible with imperfect information flows or heterogeneous reactions in beliefs across the two regions. First, we split our sample between firms that export to the US and those that do not. Consistent with our identifying assumption, we find no difference in outcomes across these two segments. Second, we use surveys from professional forecasters regarding cash flow growth expectations to examine whether investors based in Mainland China adjust their cash flow forecasts differently than investors based in Hong Kong.³ In line with our identifying assumption, monetary policy surprises do not result in heterogeneous revisions in cash flow expectations from analysts across the two regions. In further robustness exercises, we also find no significant reaction in exchange rates between the Chinese yuan (CNY) and HKD or USD following a monetary policy surprise, which rules out exchange rate dynamics as an alternative channel. Our results are also robust to alternative clustering, fixed effects, time periods, and sample restrictions, as well as a series of alternative specifications.

¹ To our knowledge, no prior study has successfully isolated the transmission of monetary policy into stock prices specifically via the discount factor channel. Work such as Hanson and Stein (2015) and Hanson et al. (2021) has studied the reaction of bond discount factors to monetary surprises and documented the existence of an “excess reaction puzzle”. Stock discount factors are, however, distinct from those of bonds as shown by Fama and French (1993) and more recently by Cieslak and Pang (2021).

² For example, Eugene Fama noted in 2020 that “The central banks don't do anything real... we don't see a big effect of Fed actions on real activity or on stock prices or on anything else”.

³ Jia et al. (2017) find that A-shares exhibit a notably stronger reaction to recommendation adjustments by local analysts as opposed to foreign analysts.

The remainder of the paper is organized as follows. Section 2 presents our empirical strategy to identify the effects of monetary policy on asset prices, and Section 3 describes the data. Section 4 analyzes the empirical results, Section 5 discusses our robustness tests of the results, and Section 6 concludes.

2. Methodology

In this section, we discuss the institutional setting and outline our empirical approach to capture the effect of monetary policy on equity prices through discount factors. We achieve this result using shares for the same company that are listed in segmented markets: Mainland China and Hong Kong.

2.1. Institutional setting

Dual listed shares. Mainland China and Hong Kong maintain distinct stock exchanges, each shaped by different historical trajectories. The Stock Exchange of Hong Kong Limited (SEHK), established in 1891, serves as the primary exchange in Hong Kong. In contrast, Mainland China operates two major exchanges – the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE) – both founded in 1990. Since their inception, these mainland exchanges have expanded rapidly, with a significant increase in the number of listed stocks. Mainland exchanges issue two types of shares: A-shares and B-shares. The vast majority of firms (5048 out of 5130) have issued A-shares, which are denominated in CNY and restricted to domestic investors. A small number of firms (82 out of 5130) issued B-shares, which are traded in foreign currencies.⁴ B-shares are restricted to foreign investors, while A-shares are restricted to Chinese residents.⁵

Almost contemporaneously with the establishment of mainland exchanges, mainland Chinese companies also began listing in Hong Kong.⁶ Stocks issued by Chinese firms on the SEHK are commonly referred to as H-shares. The first listing of a Mainland Chinese firm in Hong Kong occurred in 1993, and since then, the number of Chinese firms on the SEHK has steadily increased. By the end of 2024, 374 Chinese firms had listed H-shares, with a combined market capitalization of \$890 billion. This figure represents approximately 19.7% of the SEHK's total market capitalization and about 8% of the total capitalization of the two Mainland China exchanges.⁷ For firms that issue both A- and H-shares, these securities represent claims on the exact same underlying cash flow stream.

The Hong Kong stock market operates with minimal capital controls, facilitating the free movement of capital. In contrast, Mainland China enforces stringent capital controls that restrict cross-border capital flows. Consequently, Mainland investors are barred from directly trading Hong Kong-listed shares, while foreign investors face significant barriers to accessing Mainland China's stock markets due to capital mobility constraints (Carpenter et al., 2021; Carpenter and Whitelaw, 2017; He et al., 2023). These restrictions have resulted in market segmentation between dual-listed shares in Hong Kong and Mainland China, leading to persistent price disparities. Specifically, arbitrage opportunities are limited, contributing to the well-documented premium of A-shares over H-shares (Jia et al., 2017). Consequently, any shocks to one stock market are likely to impact only the prices of the corresponding share and contribute to a change in the price differences. It is worth noting that investors in both markets are exposed to a similar

flow of information, given the cultural and geographical proximity, and endowed with equal voting and cash-flow rights.⁸ The A/H premium fluctuates over time, with A-shares occasionally trading at double the price of their H-share counterparts. Fig. 1 illustrates the variation in the A/H premium over time.

Monetary regimes in Hong Kong and mainland China. Despite the fact that Hong Kong was returned to China in 1997, the former British territory has maintained an independent financial system that differs from that of Mainland China. Hong Kong and Mainland China maintain different currencies and operate separate monetary policy. The unique institutional setting – which features significant capital controls and different central banks that affect different shares of the same company – provides an ideal setting for identifying the transmission of monetary policies through discount rates.

Hong Kong has a separate currency, the HKD, which is pegged to the USD. The Hong Kong Monetary Authority (HKMA) implements the Linked Exchange Rate System (LERS) in Hong Kong, whereby the exchange rate between HKD and USD is pegged within a range of 7.75 to 7.85 HKD per USD. To uphold the fixed exchange rate, the monetary base of Hong Kong is fully backed by USD assets, and the HKMA provides Convertibility Undertakings (CUs), under which the HKMA commits to selling HKD to banks at 7.75 HKD per USD and buying HKD from banks at 7.85 HKD per USD. The policy rate in Hong Kong is pegged to the target rate in the US.⁹

Mainland China, on the other hand, uses the CNY and operates an independent monetary policy. The People's Bank of China (PBoC) employs various tools to ensure that monetary policy aligns with the economic challenges faced by Mainland China. Over time, the Chinese exchange rate system has become increasingly flexible. In 2005, the PBoC introduced a new exchange rate regime called “managed float-ing”. During this period, the PBoC still heavily intervened in the foreign exchange market, which resulted in a relatively stable rate compared with the USD. We thus see little effect of US monetary policy surprises on differences in exchange rates between the HKD and CNY. Since July 2015, the PBoC has pursued and maintained a more flexible exchange rate regime.

2.2. Conceptual framework

To clarify our empirical approach, consider the following conceptual framework: The economy is characterized by integrated trade but fully segmented financial markets between two regions, *A* and *H*, populated by different investors. The economy features a continuum of firms whose stock shares are divided and sold in parts in the two regions. Shares are claims on firms' future cash flow,¹⁰ which are

⁴ Data are from Shanghai Stock Exchange (2025) and Shenzhen Stock Exchange (2025).

⁵ For more details, see Carpenter et al. (2020); Jia et al. (2017); and Mei et al. (2009).

⁶ Many Chinese firms have also listed their stocks in other global financial centers such as New York, London, and Singapore.

⁷ Data are from Stock Exchange of Hong Kong Limited (2025), Shanghai Stock Exchange (2025) and Shenzhen Stock Exchange (2025).

⁸ The market was partially connected with the introduction of the Shanghai-Hong Kong Stock Connect in November 2014. Under the program, a limited number of stocks were allowed to be traded by investors in both Hong Kong and Mainland with more than RMB500,000 (or approximately \$70,000 USD) in their accounts. The introduction of Stock Connect is discussed by He et al. (2023). Suggesting the persistence of de facto segmentation with several restrictions in access, the introduction of the Stock Connect did not lead to significant reductions in the A/H premium. To ensure our results are not affected by the Stock Connect, we nonetheless also run our regressions on a restricted sample consisting of the period pre-2014 when the markets were more strictly segmented and find similar results.

⁹ From November 1998 to October 2008, the policy rate in Hong Kong was set as either the target rate in the US plus 150 bp or the simple average of the 5-day moving averages of the overnight and 1-month HIBOR, whichever was higher. After October 2008, the premium of the Hong Kong policy rate over the US target rate was reduced to 50 basis points.

¹⁰ Note that, for expositional simplicity, we assume a single cash flow. Assuming additional cash flows would complicate the relation between the ratio of prices and the ratio of discount factors so that the expression is to be interpreted as an approximation. Additionally, we abstract from other potential determinants of the A/H premium.

identical for all shares of the same firm, irrespective of the market it is trading in. Each region follows its own monetary policy stance—respectively, m_A and m_H . In line with the monetary policy surprises literature, a monetary policy stance is to be interpreted as the reaction function for the central bank across all policy instruments to every relevant event.

Surprises to the monetary policy stance, as revealed through monetary policy announcements, may affect stock prices directly through firms' cash flows (aggregate demand channel) and indirectly by affecting investors' risk-adjusted discount factor (discount factor channel). While the monetary policy stance can affect cash flows in both regions through integrated trades, it only influences the discount factor of its own region's investors due to capital controls. Following these assumptions, we write the price of the stock of company i traded in both regions A and H following [Cochrane \(2005\)](#) as the future expected discounted cash flows:

$$P_A^i = \frac{1}{R_A^i(m_A)} E[x^i(m_A, m_H)], \quad P_H^i = \frac{1}{R_H^i(m_H)} E[x^i(m_A, m_H)], \quad (1)$$

where P_J^i is the price of stock i in region $J \in \{A, H\}$, R_J^i is the risk-adjusted discount factor applied to stock i 's cash flow by investors in region J , and $E[x^i(m_A, m_H)]$ is the cash flow expectation of stock i which, unlike the discount factors, possibly depends on the monetary policy stance in both regions. We further decompose the discount factor into a risk-free discount rate and a risk premium element:

$$R_J^i = r^f(m_J) + r^{p,i}(m_J), \quad \text{for } J \in \{A, H\}.$$

This decomposition explicitly highlights that the monetary policy stance can influence the discount factor through both the risk-free rate r^f and the firm-specific risk premium $r^{p,i}$. Taking the ratio of stock prices across the two regions:

$$\frac{P_A^i}{P_H^i} = \frac{R_H^i(m_H)}{R_A^i(m_A)}. \quad (2)$$

That is, the dependence of stock prices on monetary policy through cash flows cancels out because the ratio of stock prices across the two regions is proportional to the sum of the ratios of the discount factors in the two regions. This equation motivates our empirical approach because focusing on this ratio around monetary policy announcements in a given region allows us to extract the effect of surprises to the monetary policy stance on stock prices that are due to changes in relative discount factors. For instance, if a surprise to the monetary policy stance m_H causes region H investors to lower the risk premium they require for holding stock i , $r^{p,i}$, or their time-discount rate, r^f , it will result in a lower discount factor R_H^i and, consequently, a decrease in the price ratio P_A^i/P_H^i .

2.3. Empirical strategy

Following the above conceptual framework, our empirical strategy exploits the same stocks listed simultaneously in Mainland China and Hong Kong during monetary policy announcements. Focusing on the ratio of share prices for the same company allows us to remove the effect of cash-flow revisions on prices, since the A- and H-securities are claims to the same company's cash flow and thus isolate discount rate responses to monetary policy announcements. In our main analysis, we estimate variants of the following specification:

$$(P_A/P_H)_{ist} = \alpha_i + \eta_s + \lambda_t + \beta \text{Surprise}_s \times \text{Post}_t + \varepsilon_{ist}. \quad (3)$$

In this regression equation, $(P_A/P_H)_{ist}$ is the A/H ratio of company i stock prices listed in Mainland China (P_A) and Hong Kong (P_H), around announcement s in event time t . Surprise_s is the surprise component of announcement s . Post_t is an indicator variable equal to one if event time t is after the relevant announcement s and zero otherwise. We also include firm fixed effects α_i , which absorb time-invariant company-specific factors such as firm size or location; announcement fixed effects



Fig. 1. Time series of median A/H ratio.

This figure plots the daily time series of the median A/H share price ratio from December 2001 to October 2024.

η_s , which absorb announcement-specific factors such as macroeconomic conditions; and event time fixed effects λ_t , which absorb aggregate temporal effects. Our main coefficient of interest is β , which captures the effect of a one percentage point surprise announcement on the ratio of share prices in Mainland China and Hong Kong.¹¹

Our primary identification assumption is that, in the absence of monetary policy announcements, A- and H-shares would have trended smoothly. This assumption relies on the fact that the two share types are claims to the *exact same cash flows*. Thus in the absence of any change in discount rates, we should not see significant shifts in the ratio of A- and H-shares. The inclusion of fixed effects is important to our strategy as the well-documented level differences in the A- and H-shares—which may be driven by differences in demand between Mainland China and the outside world or other factors—are absorbed by company fixed effects α_i . Therefore, to threaten our empirical approach, a confounding factor must not only be correlated with the discount rate of either only A- or H-shares but also occur at the time of monetary policy announcements.

In this regard, a possible challenge to identification is that beliefs about future cash flows react differently in each region. Note that, in order for this divergence to be a threat to our identification, investors in Hong Kong and Mainland China would have to perceive US monetary policy surprises as having differential effects on firms' cash flows. This assumption is weaker than the need for beliefs to be identical in Mainland China and Hong Kong. If Mainland Chinese are, on average, more optimistic or pessimistic, this would be absorbed by the company fixed effects α_i . In [Appendix B](#), we present evidence that analysts' beliefs are similar and react similarly to monetary policy announcements in Hong Kong and Mainland China.¹²

Throughout, we further provide graphical representations for our event study models. Doing so helps in evaluating our identification assumptions, as it demonstrates smooth trends prior to surprise announcements and allows us to study the dynamics of policy surprises. Specifically, when exploring the impact of monetary policy on bank lending rates, our analysis is at the announcement level. To assess whether the HIBOR and HIBOR futures respond to US monetary policy

¹¹ A potential concern with the approach is that heterogeneous and dynamic effects can bias estimates, which is the focus of recent literature ([Baker et al., 2022](#)). In [Appendix D](#), we address these concerns by employing stacked estimators that are robust to these effects.

¹² Another possible concern is that exchange rates may shift. Although unlikely, given that both the HKD and the CNY are stabilized against the USD, we show in [Appendix C](#) that exchange rates do not change following announcements.

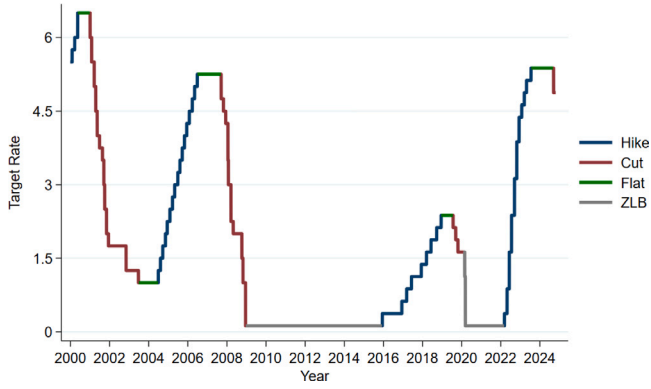


Fig. 2. Interest rate cycle.

This figure plots the target rate over time. The blue segments represent interest rate hike periods, the red segments represent interest rate cut periods, the green segments represent interest rate flat periods, and the gray segments represent the Zero Lower Bound periods. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

surprises, we use the following specification:

$$HIBOR_{st} = \eta_s + \lambda_t + \sum_{\tau=-4}^5 \beta_{\tau}^{HB} Surprise_s \times \mathbb{1}(Time\ Since\ Announcement_{st} = \tau) + \varepsilon_{st}. \quad (4)$$

Here, $HIBOR_{st}$ is the 3-month HIBOR or 3-month HIBOR future around announcement s at event time t and $\mathbb{1}(Time\ Since\ Announcement_{st} = \tau)$ is an indicator variable that equals one if the time since the relevant announcement s equals τ at event time t and zero otherwise. Other variables are the same as those in Eq. (3). In most specifications, we include announcement fixed effects η_s as well as event time fixed effects λ_t . The coefficients β_{τ}^{HB} capture the difference in rates following a monetary policy surprise in a specific period τ .

We conduct similar analysis at the company-announcement level on the ratio share prices through the following specification that adds firm fixed effects α_i :

$$(P_A/P_H)_{ist} = \alpha_i + \eta_s + \lambda_t + \sum_{\tau=-4}^5 \beta_{\tau} Surprise_s \times \mathbb{1}(Time\ Since\ Announcement_{st} = \tau) + \varepsilon_{ist}. \quad (5)$$

In Eq. (5), all variables are the same as those defined in Eqs. (3) and (4), and we are again interested in the coefficients β_{τ} that capture the difference between A- and H-shares following a monetary policy surprise. We cluster standard errors at the company level.¹³

3. Data

This section describes the main data sources for share prices, rates, and policy surprises. Our main data sources are Wind Information, the HKMA, and the Federal Reserve System.

3.1. Share prices

Our stock price data primarily come from Wind Information, a company that specializes in financial software services and data provision in China. Wind has built a large database on Chinese financial and securities data, which includes share prices for publicly listed companies.

¹³ In Online Appendix E, we show that results are robust to alternative, more stringent, clustering.

Table 1
Summary statistics.

	Mean	SD	25th Pct.	Median	75th Pct.
Panel A: Company-announcement level					
A	15.99	40.25	3.884	7.058	14.55
ΔA	-0.0276	1.106	-0.0864	-0.00365	0.0663
H	8.354	16.47	1.892	3.608	8.096
ΔH	-0.000765	0.780	-0.0530	-0.00203	0.0437
A/H	2.339	1.882	1.415	1.818	2.663
$\Delta A/H$	-0.00637	0.116	-0.0278	-0.000424	0.0229
N	8496				
Panel B: Announcement level					
Target rate	2.878	1.862	1.250	2.250	4.875
Δ Target rate	0.00291	0.286	0	0	0.250
Surprise	-0.0151	0.0929	-0.00999	0	0.00484
3-Month HIBOR	2.683	1.701	1.170	2.261	4.209
Δ 3-Month HIBOR	-0.0154	0.103	-0.0301	0	0.0236
3-Month HIBOR future	2.732	1.684	1.250	2.250	4.250
Δ 3-Month HIBOR future	-0.00574	0.107	-0.0400	0	0.0300
N	129				

This table presents summary statistics for our main analysis variables on announcement dates. All announcements listed in Table 2 are included in the sample. Panel A summarizes the data at the company-announcement level. A , H , and A/H , respectively, represent the A-share price, H-share price, and A/H ratio on the announcement date. ΔA , ΔH , and $\Delta A/H$, respectively, represent the difference in A-share prices, H-share prices, and A/H ratios between the announcement date and the day before the announcement date. Note that the sample includes both increases and decreases. Panel B summarizes data at the announcement level. *Target Rate*, *3-Month HIBOR*, and *3-Month HIBOR Future*, respectively, represent the target rate, 3-month HIBOR, and 1-month future rate of 3-month HIBOR on the announcement date. Δ Target Rate, Δ 3-Month HIBOR, and Δ 3-Month HIBOR Future, respectively, represent the difference in target rates, 3-month HIBOR, and 1-month future rates of 3-month HIBOR between the announcement date and the day before the announcement date. *Surprise* represents the surprise component of the change in target rate.

Throughout the period from June 2000 to September 2024, a total of 143 dual-listed stocks were traded both in Mainland China and Hong Kong stock markets. We collect daily closing stock prices for A-H share pairs over time. In parallel, we obtain the HKD-CNY exchange rate from Wind to ensure currency consistency. For our baseline analysis, all stock prices are converted to CNY. For each dual-listed stock, we merge its daily price data with the set of FOMC announcements, retaining a window of five consecutive trading days before and after each announcement date. Finally, we combine all dual-listed stocks to construct our sample for the event studies. Table 1 presents the summary statistics of our sample on announcement dates. In Panel A, we present stock-event-level statistics, which include A-share and H-share prices, as well as the A/H ratio. In Panel B, we present event-level statistics, which include target rates, surprises, HIBOR, and HIBOR futures.

3.2. Interbank rates

We collect data on relevant interest rates from the HKMA, Datastream, and Bloomberg. The primary rate of interest is the HIBOR, which represents the annualized rate for interbank lending on instruments denominated in HKD. The Hong Kong Association of Banks calculates and publishes HIBOR on each working day. We retrieve historical daily data from the HKMA (Hong Kong Monetary Authority, 2025). HIBOR is available across multiple maturities, including overnight, 1 week, 1 month, 3 months, 6 months, 9 months, and 1 year. For our baseline analysis, we primarily focus on the 3-month HIBOR, as its futures data offer a longer time-series history. We obtain daily data on 1-month futures of the 3-month HIBOR from Datastream. Our dataset on HIBOR and its futures spans from June 2000 to September 2024.

We also collect data on the London Interbank Offered Rate (LIBOR) and the Shanghai Interbank Offered Rate (SHIBOR), both of which

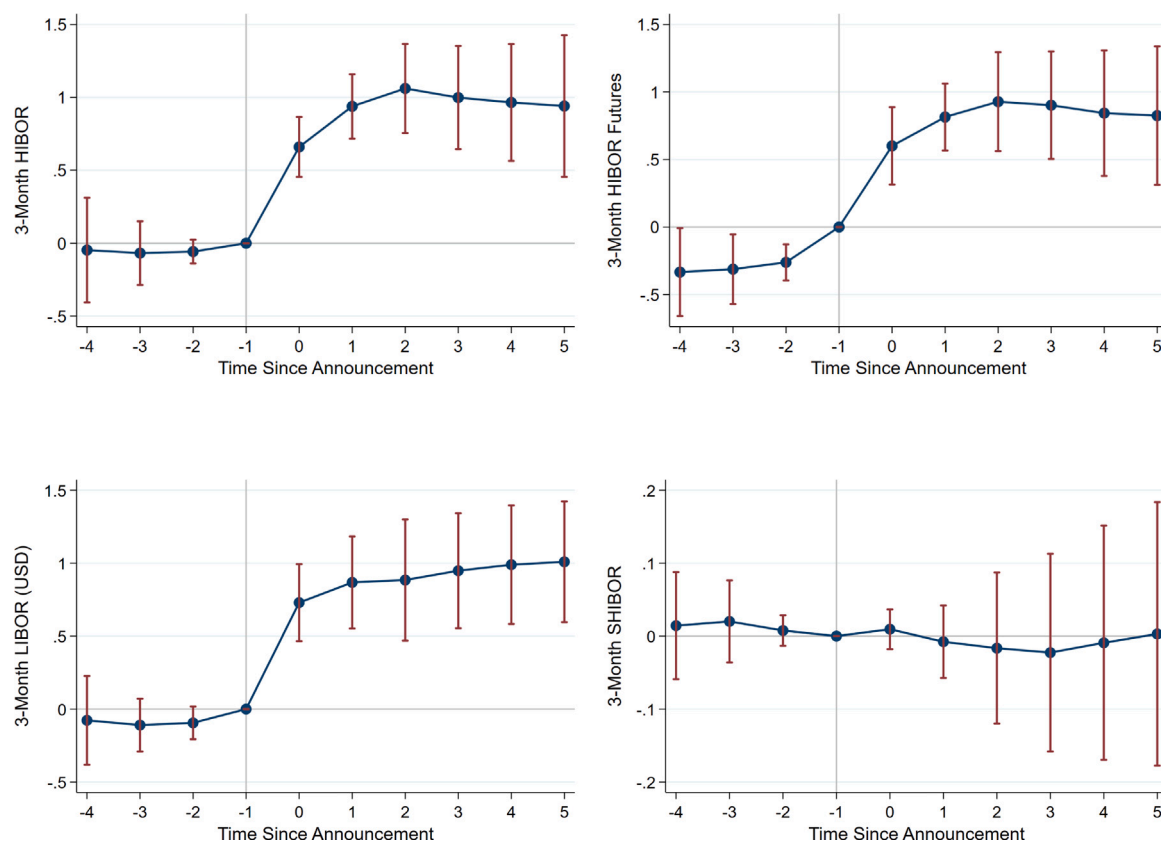


Fig. 3. Policy announcements, HIBOR, LIBOR, and SHIBOR.

This figure plots estimates of Eq. (4) around FOMC announcements, where the dependent variable is HIBOR, HIBOR futures, LIBOR in USD, or SHIBOR, all of which have a 3-month maturity. For HIBOR and HIBOR futures, all announcements listed in Table 2 are included in the sample. For LIBOR, announcements up to June 14, 2023 in Table 2 are included in the sample given the discontinuation of LIBOR in summer 2023. For SHIBOR, announcements after October 25, 2006 in Table 2 are included in the sample because SHIBOR launched in October 2006. The vertical axis depicts HIBOR, HIBOR futures, LIBOR in USD, or SHIBOR. Announcement fixed effects and a post-announcement indicator are included. Standard errors are clustered at the announcement level. In the top left panel, the dependent variable is HIBOR. In the top right panel, the dependent variable is HIBOR futures. In the bottom left panel, the dependent variable is LIBOR in USD. In the bottom right panel, the dependent variable is SHIBOR. The horizontal axis is the event-time indicator. The time since the announcement equals zero if the FOMC announcement was published on that date. Blue dots represent the regression coefficients, and red bands represent 95% confidence intervals. The vertical gray line marks time since announcement = -1, where we normalize the coefficients to zero. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

represent annualized interbank lending rates. LIBOR is denominated in various currencies, including GBP and USD; given our focus on US monetary policy, we use the USD-denominated LIBOR. SHIBOR, in contrast, is denominated in CNY. Both LIBOR and SHIBOR are available across multiple maturities, and for consistency, we focus on the 3-month rate. Historical daily LIBOR data are sourced from Bloomberg, while SHIBOR data are retrieved from the official website ([Shanghai Interbank Offered Rate, 2024](#)). Our LIBOR dataset spans June 2000 to July 2023, reflecting the rate's discontinuation in mid-2023. The SHIBOR dataset covers the period from October 2006 – its launch date on October 8, 2006 – through September 2024. We finally merge the HIBOR, HIBOR futures, LIBOR, and SHIBOR data with the event set.

3.3. Policy surprises

To identify the surprise component of each FOMC announcement, we follow [Kuttner \(2001\)](#). We obtain data on FOMC announcement dates, target rates, and federal funds futures.

Our dataset includes FOMC announcements from June 2000 to September 2024, and the announcement dates are obtained from the

FOMC's website.¹⁴ Target rate data is sourced from the Federal Reserve Economic Data (FRED). Before December 15, 2008, the Fed set a single target rate, which we use directly. From December 16, 2008, onward, the Fed established an upper and lower bound; we take their average as the target rate. We obtain the spot and 1-month federal funds futures rates from Datastream. After computing the surprise components, we merge the announcement dates, target rates, and surprises to construct our event set.

In our baseline analysis, we exclude FOMC announcements from December 16, 2008, to December 16, 2015, during which the federal funds target rate was maintained at the zero lower bound in response to the financial crisis. Similarly, we omit announcements from March 3, 2020, to March 15, 2022, when the target rate was again set to zero amid the economic disruptions caused by the COVID-19 pandemic. These exclusions aim to mitigate potential confounding effects associated with periods of unconventional monetary policy. Nonetheless, our main results remain robust when these announcements are included in the analysis. Table 2 presents a comprehensive list of events in the full

¹⁴ We exclude two stocks that became dual-listed after the FOMC meeting on September 18, 2024.

Table 2
List of FOMC announcements.

Announcement date	Target rate	Kuttner surprise	Announcement date	Target rate	Kuttner surprise	Announcement date	Target rate	Kuttner surprise
2000/06/28	6.5	-.02	2005/08/09	3.5	0	2017/06/14	1.125	.009
2000/08/22	6.5	-.017	2005/09/20	3.75	.015	2017/07/26	1.125	0
2000/10/03	6.5	0	2005/11/01	4	0	2017/09/20	1.125	0
2000/11/15	6.5	0	2005/12/13	4.25	0	2017/11/01	1.125	0
2000/12/19	6.5	.052	2006/01/31	4.5	0	2017/12/13	1.375	.004
2001/01/03	6	-.382	2006/03/28	4.75	0	2018/01/31	1.375	-.005
2001/01/31	5.5	.005	2006/05/10	5	-.007	2018/03/21	1.625	.008
2001/03/20	5	.056	2006/06/29	5.25	-.015	2018/05/02	1.625	0
2001/04/18	4.5	-.425	2006/08/08	5.25	-.04	2018/06/13	1.875	-.004
2001/05/15	4	-.078	2006/09/20	5.25	0	2018/08/01	1.875	-.003
2001/06/27	3.75	.05	2006/10/25	5.25	0	2018/09/26	2.125	0
2001/08/21	3.5	.016	2006/12/12	5.25	0	2018/11/08	2.125	.003
2001/10/02	2.5	-.069	2007/01/31	5.25	0	2018/12/19	2.375	.032
2001/11/06	2	-.1	2007/03/21	5.25	0	2019/01/30	2.375	0
2001/12/11	1.75	0	2007/05/09	5.25	0	2019/03/20	2.375	.014
2002/01/30	1.75	.015	2007/06/28	5.25	0	2019/05/01	2.375	-.021
2002/03/19	1.75	-.026	2007/08/07	5.25	.026	2019/06/19	2.375	.041
2002/05/07	1.75	0	2007/08/17	5.25	.155	2019/07/31	2.125	.045
2002/06/26	1.75	0	2007/09/18	4.75	-.15	2019/09/18	1.875	-.033
2002/08/13	1.75	.034	2007/10/31	4.5	-.02	2019/10/30	1.625	-.017
2002/09/24	1.75	.025	2007/12/11	4.25	.008	2019/12/11	1.625	0
2002/11/06	1.25	-.194	2008/01/22	3.5	-.741	2020/01/29	1.625	.015
2002/12/10	1.25	0	2008/01/30	3	-.095	2022/03/16	.375	-.027
2003/01/29	1.25	.005	2008/03/18	2.25	.167	2022/05/04	.875	-.015
2003/03/18	1.25	.048	2008/04/30	2	-.055	2022/06/15	1.625	.036
2003/05/06	1.25	.037	2008/06/25	2	-.03	2022/07/27	2.375	-.031
2003/06/25	1	.15	2008/08/05	2	-.006	2022/09/21	3.125	-.043
2003/08/12	1	0	2008/09/16	2	.059	2022/11/02	3.875	-.002
2003/09/16	1	0	2008/10/08	1.5	-.142	2022/12/14	4.375	0
2003/10/28	1	0	2008/10/29	1	-.06	2023/02/01	4.625	-.008
2003/12/09	1	0	2008/12/16	.125	-.119	2023/03/22	4.875	.045
2004/01/28	1	0	2015/12/16	.375	.021	2023/05/03	5.125	.028
2004/03/16	1	0	2016/01/27	.375	0	2023/06/14	5.125	-.017
2004/05/04	1	-.006	2016/03/16	.375	-.01	2023/07/26	5.375	0
2004/06/30	1.25	-.01	2016/04/27	.375	0	2023/09/20	5.375	-.015
2004/08/10	1.5	.022	2016/06/15	.375	0	2023/11/01	5.375	0
2004/09/21	1.75	.017	2016/07/27	.375	0	2023/12/13	5.375	-.003
2004/11/10	2	0	2016/09/21	.375	-.042	2024/01/31	5.375	-.003
2004/12/14	2.25	0	2016/11/02	.375	-.005	2024/03/20	5.375	0
2005/02/02	2.5	0	2016/12/14	.625	0	2024/05/01	5.375	0
2005/03/22	2.75	0	2017/02/01	.625	-.003	2024/06/12	5.375	0
2005/05/03	3	0	2017/03/15	.875	.005	2024/07/31	5.375	.01
2005/06/30	3.25	0	2017/05/03	.875	0	2024/09/18	4.875	-.106

This table lists the events that are used in the baseline specifications. *Announcement Date* is the date of FOMC announcements. *Target Rate* is the target rate specified in FOMC announcements. After December 2008, it is calculated as the average of the upper bound and the lower bound. *Kuttner Surprise* is the surprise component of the changes in the target rate calculated according to Kuttner (2001).

sample while Fig. 2 depicts the entire monetary policy cycle for our sample and a decomposition into hiking, cutting and flat phases used in some sample-splitting exercises below.

4. Results

This section presents our main findings. As hypothesized, US monetary policy surprises lead to an almost one-to-one change in interbank rates in Hong Kong, while having no measurable effect on interbank rates in Mainland China. We then report the results of our main event-study regression, which reveal a strong impact of monetary policy surprises on the A/H ratio. Finally, we contextualize these findings within the existing literature and explore additional cross-sectional patterns.

4.1. Interbank rates

We begin by analyzing the pass-through of US monetary policy surprises to various interbank rates to assess the validity of our empirical framework. First, we examine their impact on Hong Kong's main interbank lending rate, the HIBOR. The top panel of Fig. 3 presents coefficient estimates from Eq. (4) with a 95% confidence interval. The top left panel illustrates the 3-month HIBOR rate, while the top right

panel shows the 3-month HIBOR futures. These figures indicate little to no pre-trend, followed by a sharp increase in the regression coefficients. A one percentage point surprise increase (decrease) results in an approximately one-point increase (decrease) in the 3-month HIBOR rate, with a slightly smaller effect on 3-month HIBOR futures. However, by the following day, we cannot reject the null hypothesis of a one-to-one pass-through.

The bottom left panel of Fig. 3 illustrates the reaction of LIBOR to the same monetary policy announcements for reference.¹⁵ The bottom right panel depicts the response of Mainland China's main interbank rate, the SHIBOR. As expected, the results provide strong evidence of a significant pass-through of US monetary policy to Hong Kong's financial conditions, while Chinese monetary policy remains unaffected—consistent with the presence of capital controls and market segmentation.

4.2. Monetary policy surprises and the A/H ratio

Table 3 presents our main findings from estimating Eq. (3). Column (1) shows a baseline specification without company fixed effects

¹⁵ Since LIBOR was discontinued in the summer of 2023, we restrict our sample to announcements up to June 14, 2023, as listed in Table 2.

Table 3
Main results.

	(1)	(2)	(3)	(4)	(5)
Surprise $\times$ Post	0.306*** (0.0298)	0.292*** (0.0296)	0.293*** (0.0297)	0.292*** (0.0298)	0.276*** (0.0314)
Company FE	No	No	Yes	Yes	Yes
Event time	No	Post	Post	FE	FE
Company Spec. Trend	No	No	No	Common	Separated
Announcements	129	129	129	129	129
Observations	93,414	93,414	93,414	93,414	93,414

This table presents the baseline results of estimating Eq. (3). All announcements listed in Table 2 are included in the sample. In each specification, we control for announcement fixed effects. *Company FE* represents whether we control for company fixed effects. *Event Time* indicates how event time effects are controlled for: *No* indicates that event time is not controlled for; *Post* indicates that a dummy variable for post-announcement event time is included; *FE* indicates that event time fixed effects are included. *Company Spec. Trend* represents whether and how company-specific time trends are controlled for: *No* indicates company-specific time trends are not included; *Common* indicates that we control for company-specific time trends common to surprise increases and decreases; *Separated* indicates that we control for company-specific time trends separated for surprise increases and decreases. *Announcements* represents the number of FOMC announcements. *Observations* represents the number of observations. Standard errors clustered at the company level are in parentheses. *** represents significance at the 1% level, ** 5% level, * 10% level.

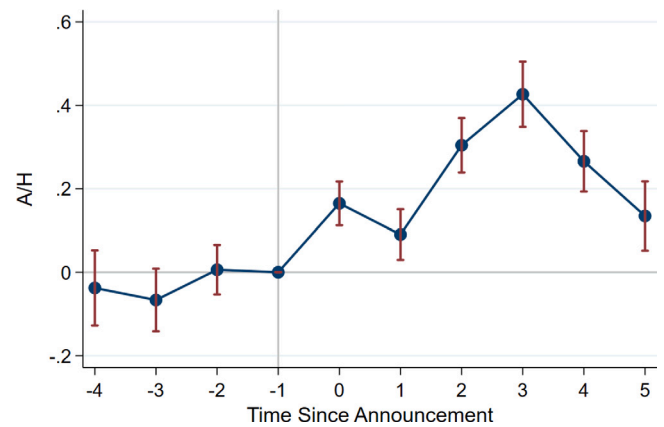
Table 4
Main results split by surprise direction and interest rate cycle.

	(1) Increase	(2) Decrease	(3) Near zero
Panel A: Interest rate hike period			
Surprise $\times$ Post	0.390*** (0.112)	−0.0964 (0.107)	−0.459 (0.439)
Announcements	8	8	38
Observations	7535	9790	31,822
Panel B: Interest rate cut period			
Surprise $\times$ Post	0.0467 (0.179)	0.355*** (0.0400)	2.435 (3.726)
Announcements	13	19	9
Observations	5906	10,541	3882
Panel C: Interest rate flat period			
Surprise $\times$ Post	−0.351 (0.228)	−0.521 (0.528)	−0.630 (0.702)
Announcements	5	5	24
Observations	3575	3419	16,944

This table presents the results of estimating variants of Eq. (3) split by both interest rate cycle and surprise changes. Panel A employs the FOMC announcements during interest rate hike periods, Panel B during interest rate cut periods, and Panel C during interest rate flat periods. The interest rate cycle is defined following Fig. 2. Column (1) presents the result using the announcements in Table 2 with surprise increases ($Surprise \geq 0.01$), Column (2) presents the result using the announcements in Table 2 with surprise decreases ($Surprise \leq -0.01$), and Column (3) presents the result for announcements in Table 2 with near-zero surprises ($-0.01 < Surprise < 0.01$). In each specification, we control for announcement fixed effects. *Announcements* represents the number of FOMC announcements. *Observations* represents the number of observations. Standard errors clustered at the company level are in parentheses. *** represents significance at the 1% level, ** 5% level, * 10% level.

or event-time controls. In Column (2), we include a dummy variable for the post-announcement period (“Post”) but still do not use company fixed effects. Columns (3) through (5) then add company fixed effects but differ in how they handle event-time controls.¹⁶ Across all specifications, the coefficient on the monetary policy surprise variable remains highly significant and stable, indicating that a one-percentage-point

¹⁶ Column (3) uses a post-announcement dummy, while Columns (4) and (5) incorporate event-time fixed effects—and in whether company-specific trends are included (none in (3), common in (4), and separated in (5)).

**Fig. 4.** Policy announcements and share prices.

This figure plots estimates of Eq. (5) around FOMC announcements, where the dependent variable is the A/H share price ratio. All announcements listed in Table 2 are included in the sample. Company and announcement fixed effects, as well as a post-announcement indicator, are included. Standard errors are clustered at the company level. The horizontal axis is the event-time indicator. The time since the announcement equals zero if the FOMC announcement was published on that date. Blue dots represent the regression coefficients, and red bands represent 95% confidence intervals. The vertical gray line marks time since announcement = −1, where we normalize the coefficients to zero. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

surprise decrease in policy rates is associated with a 27-to-30-basis-point reduction in the ratio between share prices in Mainland China and Hong Kong.

In Table 4, we reproduce our main result when splitting the sample between surprises associated with rate hikes and cuts according to the phase of the monetary cycle during which the FOMC announcement occurred. Panel A reproduces our main exercise exclusively for interest rate hike phases and reports the estimate separately for surprise increase, decrease and near-zero; Panel B and C perform the same exercise for phases of the monetary policy cycle in which interest rates are decreasing or flat, respectively. From this table, we observe that both surprise hikes during hiking cycles and surprise cuts during cutting cycles are economically and statistically significant with a coefficient of 0.390 and 0.355, respectively, both close to our main estimate of 0.30. In contrast, all other configurations, including surprise hikes during cutting cycles and surprise cuts during hiking cycles, appear statistically insignificant. Those results suggest that investors are particularly reactive to surprises that amplify the current phase of the cycle. Although definitely establishing the causes of this pattern is beyond the scope of this article and left to future research, we note that this pattern would naturally arise in a model where investors update their expectations regarding the Fed’s propensity to intervene (Bauer et al., 2022; Cieslak and Pang, 2021; Haddad et al., 2025), particularly if those amplifying surprises provide greater information about this propensity.¹⁷

¹⁷ For completeness, Online Appendix Table E.1 presents results from re-estimating our main specification after splitting the sample by the direction of the monetary policy surprise (hikes versus cuts), without further separation by policy cycle phase. In this specification, the coefficient for surprise rate cuts is 0.344, remaining close to our main estimate of 0.30. However, the coefficient for surprise rate hikes decreases to −0.01 and becomes statistically insignificant. Further investigation reveals that this discrepancy for hike surprises is driven entirely by a single confounding observation: the emergency announcement on August 17, 2007, at the onset of the financial crisis. Excluding this single announcement restores the coefficient for surprise hikes to 0.292, which is statistically significant at the 5% level.

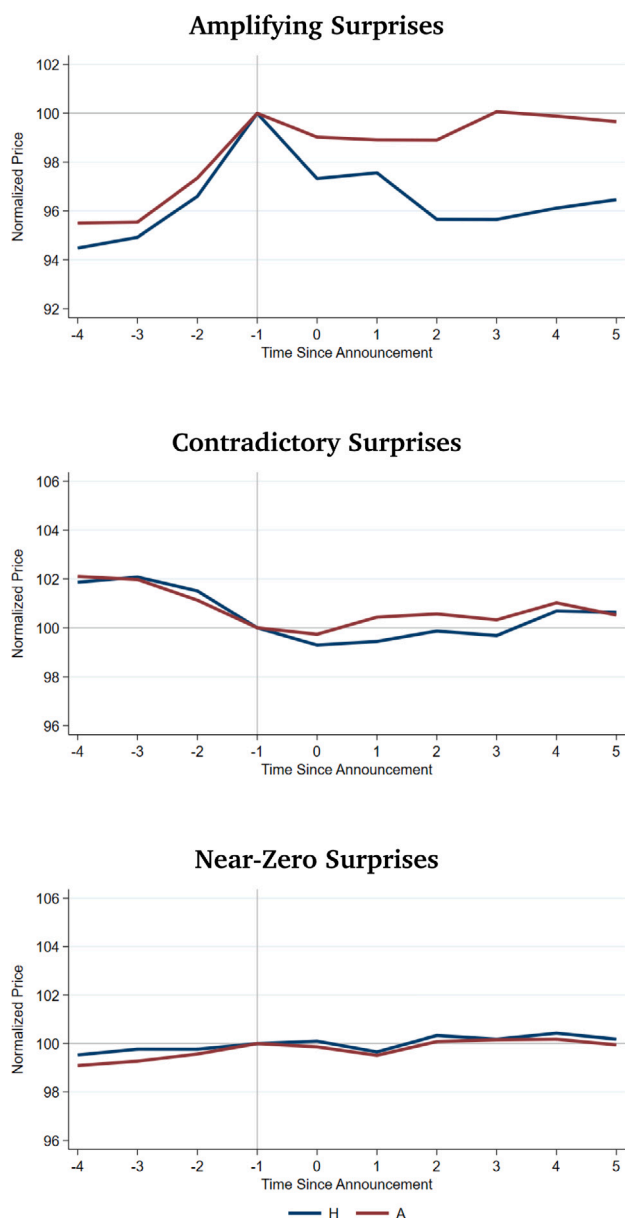


Fig. 5. Policy announcements and weighted normalized means of share prices split by surprise type.

This figure plots the weighted normalized means of H- and A-share prices around FOMC announcements with amplifying, contradictory, and near-zero surprises. The means of prices for H- (A-) shares are weighted by the absolute magnitude of surprises, and are reflected with respect to the H- (A-) share prices one day before the announcement if the announcement is a surprise decrease. The top panel shows the prices around amplifying surprises, the middle panel around contradictory surprises, and the bottom panel around near-zero surprises. Announcements with amplifying, contradictory, and near-zero surprises listed in Table 2 are included in the sample. The horizontal axis is the event-time indicator. The time since the announcement equals zero if the FOMC announcement was published on that date. The vertical axis is the weighted normalized prices. Blue lines represent H-share prices, and red lines represent A-share prices. The vertical gray line marks time since announcement = -1, where we normalize the prices for both shares to 100. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 4 displays event study estimates for a window of 4 days before and 5 days after the event, following the specification reported in column (3) of Table 3, along with 95% confidence intervals. We see no evidence of any pre-trends in the days leading up to surprise policy announcements. A positive coefficient in the post-announcement period indicates that a surprise increase (decrease) is associated with a significant increase (decrease) in the ratio between A- and H-shares. This effect appears to persist over time in the days following the announcement. In Fig. 4, the effect of the monetary policy surprise appears to intensify, peaking three days after the announcement and then mean-reverting. Taken at face-value, this result would suggest that discount factors adjust slowly and overshoot in reaction to monetary policy surprises.¹⁸ While this increase-and-reversal pattern is statistically significant when standard errors are clustered at the company level (as is customary in this context), we exercise caution in drawing strong conclusions from these specific patterns. Notably, when applying the bootstrap procedure introduced by Hounyo and Lin (2024) to compute standard errors (see Online Appendix Figure E.8) that accounts for serial correlation, the confidence intervals across event-times overlap.¹⁹ Motivated by the asymmetrical results of Table 4, Online Appendix Figure E.9 graphically decomposes, in further detail, the dynamic response of the A-H ratio following monetary policy surprises according to the concordance between the direction of the surprise and the monetary policy cycle. Consistent with the table's results, only amplifying cycles yield statistically significant effects.

We note that higher-frequency strategies that do not control for cash-flow information, such as Bernanke and Kuttner (2005) might miss these longer-term effects and thereby possibly underestimate the impact of policy announcements due to confounding information about the economy and firms' future cash flows. Although the dynamics of the effect display some mean reversion, we show in Online Appendix E that some statistically significant effect remains in a 10-day window.

Fig. 5 presents the raw information behind the event study in Fig. 4 and Table 4. Specifically, it displays the weighted normalized means of A- and H-shares before and after the policy announcements, splitting announcements into cycle-amplifying, cycle-dampening, and near-zero surprises. The means are weighted by the absolute magnitude of surprises, reflected for surprise decreases with respect to the prices one day before the announcements, and normalized to 100 one day before the announcement.²⁰ Consistent with the regression results of Table 4, the means show a decrease (increase) in Hong Kong H-share prices following a cycle-amplifying surprise rate hike (cut), while the H-share prices do not change much around contradictory and near-zero surprises. We also see little impact on mainland A-shares from FOMC announcements, which conforms with the presence of strong market segmentation and independent monetary policy in Mainland China underpinning our approach.

¹⁸ Binsbergen and Grotteria (2024) consistently find mortgage yields respond weeks later, Brooks et al. (2018) document gradual bond market drift, and Hiltenbrand (2022) show declines in long-term yields span days around FOMC announcements. Neuhierl and Weber (2024) attribute post-expansionary drifts to discount factor revisions.

¹⁹ Note that the statistical significance of our main results beyond the specific pattern of adjustments are unaffected by this more conservative clustering. Online Appendix Table E.10 further reproduces the averaging effect of Table 3 the same bootstrap procedure with all coefficients remaining statistically significant at the 1% level.

²⁰ Reflecting share prices around surprise decreases avoids price changes around surprise increases and decreases offsetting each other. Normalizing share prices allows for easier identification and comparison of trends across A- and H-shares.

Table 5

Variance decomposition: Time series variation.

	Equal-Weighted		Value-Weighted	
	Standardized (1)	Demeaned (2)	Standardized (3)	Demeaned (4)
Surprise $\times$ Post	0.185** (0.0772)	0.377** (0.157)	0.231*** (0.0534)	0.349*** (0.0806)
Surprise	-1.894 (1.442)	-3.852 (2.932)	-2.127 (1.485)	-3.211 (2.242)
Post	-0.00355 (0.00893)	-0.00721 (0.0182)	0.00216 (0.00772)	0.00326 (0.0117)
Adjusted R-squared	0.0256	0.0256	0.0324	0.0324
Announcements	129	129	129	129
Observations	1419	1419	1419	1419

This table presents time-series variance decomposition results. For each regression, we first create a time series representing the average A/H ratio across stocks, then demean this time series, and finally regress the demeaned time series on *Surprise $\times$ Post*, *Surprise*, and *Post*. All announcements in Table 2 are included in the sample. Columns (1) and (2) employ equal weight for the cross-sectional means, and Columns (3) and (4) employ value weight for the cross-sectional means. Columns (1) and (3) in addition normalized the demeaned series by the standard deviation. *Adjusted R-squared* represents the adjusted R^2 from the regressions. *Announcements* represents the number of FOMC announcements. *Observations* represents the number of observations. Standard errors clustered at the announcement level are in parentheses. *** represents significance at the 1% level, ** 5% level, * 10% level.

4.3. Discussion of the magnitudes

According to the estimates presented in Table 3, a 100-basis-point rate cut decreases the ratio of Mainland A-share prices to Hong Kong H-share prices by 0.3. Given that the observation-weighted average of this A/H ratio is 2.3 (Table 1), this change corresponds to roughly a 13% decline. In turn, a 25-basis-point cut (one-quarter of a percentage point) would imply about a 3% relative increase in H-share prices compared with A-shares. This effect is notably larger than the 1% stock price response estimated by Bernanke and Kuttner (2005) and Gürkaynak et al. (2005) for a 25-basis-point rate cut at the aggregate index level.

A series of methodological differences are possibly responsible for this discrepancy. First, Bernanke and Kuttner (2005) mostly rely on first-day effect because, within an event-study methodology, standard errors become too large beyond this time window.²¹ An advantage of our approach is that by taking the ratio of two claims on the same underlying cash flow and applying time-fixed effects, we control for the effect of new information on macroeconomic variables on cash flows and are able to expand the study's time window beyond day one. As shown in Fig. 4, the discount factor effect of monetary policy surprises remains significant five days after the announcement. In particular, we note that the day-one regression coefficient is around 0.15, which generates an increase in the stock price of H stocks relative to A stocks of 1.5% for a 25-basis-point cut, a figure closer to Bernanke and Kuttner's 1%.

Another potential explanation for the larger estimates is that our method, by isolating the impact of monetary policy through discount rates, partially accounts for the "Fed information" channel discussed in Nakamura and Steinsson (2018), Jarociński and Karadi (2020), and Hillenbrand (2022). These studies demonstrate that monetary policy surprises provide market participants with information about the state of the economy, which in turn influences expectations of firms' future cash flows. This effect often runs counter to the direct causal impact of monetary policy. For instance, a surprise interest rate hike by the Fed may signal to the market that the economy is performing better than previously anticipated. Consequently, the information channel is

²¹ Bernanke and Kuttner (2005) also report results based monthly regressions but cautioned against overinterpretation of those results due to concurrent macro-announcement releases.

often viewed as a credible explanation for why some surprise rate hikes are accompanied by an increase in stock prices (and vice versa). By focusing on the effect of monetary policy surprises on the A/H ratio, our estimates are likely less influenced by the confounding effect of the information channel.²² A larger coefficient for monetary policy surprises is consistent with this hypothesis.

To assess the explanatory power of monetary policy surprises on the average A/H ratio over time, Table 5 presents the coefficient from a time-series regression of the average A/H ratio across firms on the monetary policy surprise. The first two columns present the coefficients for the equal-weighted average A/H ratio, while Columns (3) and (4) report the coefficients for the value-weighted average A/H ratio. In Columns (1) and (3), the average ratio is standardized by dividing it by its standard deviation, while in Columns (2) and (4), the series is simply demeaned. As can be observed, the coefficients of those time-series regressions are similar to the panel data ones, suggesting that our main results are driven by changes in time-series rather than in the cross-section of shocks following a monetary policy surprise. The reported R^2 for those regressions range between 0.026 and 0.032.

4.4. Additional cross-sectional results

In this section, we explore how monetary policy surprises heterogeneously affect firms' specific discount factors across four stock characteristics in the following triple-difference:

$$\begin{aligned}
 (P_A/P_H)_{ist} = & \alpha_i + \eta_s + \lambda_t + \beta_1 \text{Surprise}_s \times \text{Post}_t \\
 & + \beta_2 \text{Charact}_{ist} + \beta_3 \text{Surprise}_s \times \text{Charact}_{ist} \\
 & + \beta_4 \text{Post}_t \times \text{Charact}_{ist} + \beta_5 \text{Surprise}_s \times \text{Post}_t \\
 & \times \text{Charact}_{ist} + \epsilon_{ist}
 \end{aligned} \quad (6)$$

where Charact_{ist} is a lagged stock characteristic of stock i on announcement s and at event time t . We first focus on three characteristics known to be significant drivers of aggregate cross-sectional variations in stock returns (Fama and French, 1992): price-to-earnings ratios (PE for value), market capitalization (MC for size), and CAPM Betas.²³ Market value data are calculated on a trading day basis following the formula $\text{Market}_{it} = H \text{Shares}_{it} \times (P_H)_{it} + A \text{Shares}_{it} \times (P_A)_{it}$, with i denoting a stock and t a trading day. $H \text{Shares}_{it}$ and $(P_H)_{it}$ denote the number of shares and the closing price in the Hong Kong stock market, while $A \text{Shares}_{it}$ and $(P_A)_{it}$ denote the number of shares and the closing price in the Mainland China stock market. The price-to-earnings ratio are $\text{Market}_{it}/\text{Earnings}_{it}$. Because prices may be affected by the monetary policy surprises, we lag our price-to-earnings ratio by a year. The CAPM betas are calculated from a 5-year rolling window based on the monthly returns of the stocks and the Shanghai Composite Index (SSE) using data from Wind.²⁴ Online Appendix Table E.2 further reports a low correlation between those characteristics, including the lagged A/H ratio.

Panel A of Table 6 displays the results of interacting *Surprise $\times$ Post* with the PE ratios. Previous studies have documented that firms with

²² While our methodology accounts for the Fed's information effect on firms' future cash flows in equity markets, a residual information effect may persist if investors adjust their long-term discount factors in response to changing economic conditions. Evidence of an information effect on long-term interest rates (r^*) is provided by Nakamura and Steinsson (2018) and Hillenbrand (2022).

²³ To compute PE ratios, MC, and CAPM betas, we retrieve total earnings, market values, and CAPM betas data from Wind, starting on June 1, 2000 for market values and CAPM betas and February 15, 2001 for total earnings.

²⁴ Specifically, the beta of stock i in month t is obtained by conducting the following regression: $r_{i,m} = \alpha_i + \beta_i r_{SSE,m} + \epsilon_{i,m}$, where $m \in \{t-60, t-59, \dots, t-1\}$ denotes a month in the rolling window of month t , $r_{i,m}$ is the monthly return of stock i in month m , and $r_{SSE,m}$ is the monthly return of SSE in month m . To alleviate the effects of outliers, we winsorize betas by 2.5% on both sides for each announcement.

Table 6
Main results interacted with characteristics.

	(1)	(2)	(3)	(4)	(5)	(6)
	Full sample	≤Median	>Median	Full sample	≤Median	>Median
Panel A: Price-to-Earnings			Panel B: Market capitalization			
Surprise × Post	0.301*** (0.0516)	0.346*** (0.0651)	0.162*** (0.0485)	2.518*** (0.529)	0.367*** (0.0575)	0.237*** (0.0346)
Charact	−0.0660 (0.0683)			−0.395*** (0.0474)		
Surprise × Charact	0.642 (0.395)			0.378*** (0.130)		
Post × Charact	−0.00825 (0.00749)			0.00238*** (0.000829)		
Surprise × Post × Charact	−0.161* (0.0908)			−0.0889*** (0.0210)		
Announcements	112	112	112	129	129	129
Observations	83,361	41,828	41,533	92,054	46,284	45,770
Panel C: CAPM β			Panel D: Lagged A/H Ratio			
Surprise × Post	−0.0368 (0.142)	0.220*** (0.0638)	0.381*** (0.0730)	0.160*** (0.0319)	0.211*** (0.0327)	0.396*** (0.0622)
Charact	0.518** (0.240)			0.0135 (0.0134)		
Surprise × Charact	−0.527 (1.373)			−0.514*** (0.0992)		
Post × Charact	−0.00337 (0.00888)			0.000156 (0.000169)		
Surprise × Post × Charact	0.362** (0.146)			0.0338*** (0.00650)		
Announcements	129	129	129	129	129	129
Observations	71,864	36,152	35,712	91,648	46,055	45,593

This table presents the results of estimating variants of Eq. (6) with the full sample and variants of Eq. (3) split by the median of the characteristic on announcement dates. In Panel A, the characteristic is price-to-earnings ratio (PE). For available firms, the PE data is extracted from Wind, is winsorized by 2.5% on both sides, and is lagged by one year. Because earlier PE data are not available, only announcements after March 19, 2002, in Table 2, are included in the sample. In Panel B, the characteristic is the log of market capitalization (MC). For available firms, the MC data is extracted from Wind and is lagged by three months. All announcements in Table 2 are included in the sample. In Panel C, the characteristic is the CAPM β . For available firms, the β of each stock is extracted from Wind, is calculated from a 5-year rolling window based on the monthly returns of the stock and Shanghai Composite Index, and is winsorized by 5% on both sides. In Panel D, the characteristic is the lagged A/H ratio. The A/H ratio of each stock is obtained from Wind, is lagged by three months, and is winsorized by 2.5% on both sides. All announcements in Table 2 are included in the sample. Columns (1) and (4) summarize the results from estimating Eq. (6) using the full sample. Columns (2) and (5) summarize the results from estimating Eq. (3) using the subsample of firms with below-median PE, MC, β , and lagged A/H ratio. Columns (3) and (6) summarize the results from estimating Eq. (3) using the subsample of firms with above-median PE, MC, β , and lagged A/H ratio. In each specification, we include announcement fixed effects and a post-announcement indicator. *Announcements* represents the number of FOMC announcements. *Observations* represents the number of observations. Standard errors clustered at the company level are in parentheses. *** represents significance at the 1% level, ** 5% level, * 10% level.

Table 7
Robustness.

	(1)	(2)	(3)	(4)
Surprise × Post	0.130*** (0.0254)	0.291*** (0.0288)	0.286*** (0.0279)	0.295*** (0.0300)
Set of announcements				
Include	ZLB Ann.			
Exclude		Zero Surprise Ann.	Holiday Ann.	Holiday Event Times
Announcements	201	58	116	116
Observations	160,885	40,766	85,741	78,030

This table presents additional robustness tests, estimating variants of Eq. (3). In each specification, we control for announcement and company fixed effects, and a post-announcement indicator is included. *Set of Announcements* depicts changes in the set of announcements and event times compared with the baseline sample. *Include* indicates the additional announcements included in the baseline sample, whereas *ZLB Ann.* indicates announcements during the Zero Lower Bound periods. *Exclude* indicates announcements or event times excluded from the baseline sample, where *Zero Surprise Ann.* means announcements with near-zero surprise ($-0.01 < \text{Surprise} < 0.01$); *Holiday Ann.* means announcements made on Mainland China or Hong Kong holidays; and *Holiday Event Times* means event times that belong to 5 consecutive trading days before and after an announcement, which are separated by a Mainland China or Hong Kong holiday from the announcement. *Announcements* represents the number of FOMC announcements employed. *Observations* represents the number of observations. Standard errors clustered at the company level are in parentheses. *** represents significance at the 1% level, ** 5% level, * 10% level.

low stock market valuations relative to measures of fundamentals, such as earnings or book equity, have outperformed the market—the so-called value anomaly (Rosenberg et al., 1985; Fama and French, 1992). It has been hypothesized that this result could be caused by pervasive behavioral biases (Lakonishok et al., 1994; De Bondt and Thaler, 1985; La Porta et al., 1997); priced risk factors, such as a distressed risk factor (Fama and French, 1995; Chan and Chen, 1991); or differences in cash flow duration (Lettau and Wachter, 2007; Binsbergen et al., 2012; Weber, 2018). Panel A of Table 6 provides consistent evidence that value stocks, as measured by firms with a low PE ratio, are more impacted by monetary policy surprises. When splitting the sample between above- and below-median PE, above-median PE firms are almost twice as impacted, with an estimated coefficient of 0.346 vs 0.162 for below-median PE firms—a difference that is statistically significant at the 1% level. The triple interaction analysis points in a similar direction, consistently showing a negative coefficient across all specifications, though with low statistical significance.

This exercise suggests that monetary policy surprises affect investors' discount factors more strongly for firms with a low market valuation relative to earnings. This result aligns with previous findings that the risk premia on short-term cash flows are, on average, larger and more volatile than on more distant ones, which results in excess volatility.²⁵ In particular, this result is consistent with the model of Lettau and Wachter (2007), in which the value premium arises as the consequence of time variations in investors' stochastic discount factors. This interpretation also corresponds with the findings for the size characteristics presented in Panel B of Table 6: Small stocks, as captured by below-median market capitalization, have stronger reactions to monetary policy shocks, which renders them more sensitive to macroeconomic business cycle news and thereby more risky.

Panel C of Table 6 presents the results of a similar triple-difference regression, this time including CAPM beta as an additional interacted variable. The analysis reveals a stronger effect for higher-beta stocks, providing evidence that monetary policy surprises influence not only the risk-free component of the discount factor decomposition but also the risk premium. This finding aligns with Savor and Wilson (2014), who show that the CAPM performs better on macroeconomic announcement days—a result later corroborated by Cieslak et al. (2019) for days within the FOMC cycle. Splitting the sample into below-median and above-median beta stocks, we observe a near doubling of the *Surprise* $\times$ *Post* coefficient, from 0.220 to 0.381.

Lastly, Panel D of Table 6 presents the results of a triple interaction with the lagged A/H ratio. As shown by both the triple interaction coefficient and the difference between the above-mean and below-mean coefficients, stocks with a higher lagged A/H ratio are more responsive to monetary policy surprises. This result is consistent with the notion that the lagged A/H ratio captures persistent stock characteristics related to the sensitivity of discount factors. These characteristics may include the three discussed earlier, as well as potentially unobserved factors.²⁶ Tables E.3, E.4, E.5, and E.6 in Online Appendix E present similar results under alternative specifications. Table E.7 further demonstrates that, among these characteristics, only lagged market capitalization strongly predicts the A/H ratio in the cross-section.

5. Alternative channels and robustness

In this section, we examine and rule out the potential alternative channels through which a US monetary policy surprise could affect the A/H ratio through revisions in cash flow expectations rather than the discount factor. We also provide evidence supporting the robustness of our main findings.

²⁵ See, for example, Binsbergen et al. (2012) and Binsbergen and Kojien (2017).

²⁶ One such unobserved characteristic could be stock liquidity and its associated exposure to a liquidity risk factor, as suggested by Pástor and Stambaugh (2003).

5.1. Expectations of future cash flows

A key potential concern for our approach is that investors in the two markets might react heterogeneously to monetary policy surprises when revising their subjective cash flow expectations. If this were true, changes in the A/H ratio could still be driven by divergences in these expectations.²⁷ We provide two sets of empirical evidence that contradict this hypothesis. First, we show that local Chinese firms, whose revenues are entirely generated within Mainland China, exhibit A/H ratio responses to US monetary policy surprises that are similar to those of exporting firms. This suggests that differences in cash flow expectations are not a primary driver of the observed effects. Second, we present direct evidence that professional forecasters' cash flow projections remain unchanged following a monetary policy surprise, further supporting our conclusion.

5.1.1. Foreign exposure

First, we merge our dataset with FactSet's GeoRev data feed to identify the share of each firm's revenue generated by exports.²⁸ This approach enables us to adjust our analysis based on firms' exposure to the US economy. If heterogeneity in subjective cash flow expectations were a significant factor driving the observed reaction in the A/H ratio, we would expect exporting firms to be more sensitive to cash flow revisions caused by changes in US monetary policy than non-exporting firms. The results of this analysis are presented in Appendix A. Columns (1) to (5) of Table A.1 indicate that the interaction term *Surprise* $\times$ *Post* $\times$ *US Export Share* is largely nonsignificant across all specifications. Columns (6) and (7) show the results of the main regression, splitting the sample between non-exporting firms (*Share* = 0) and exporting firms (*Share* > 0). These findings suggest that heterogeneous reactions to cash flow expectations are unlikely to be driving our results.

5.1.2. Analysts' forecasts

Second, to ensure our results are not driven by diverging reactions in subjective cash flow expectations, we integrate our dataset with the I/B/E/S database on professional forecasters' firm cash flow predictions. Leveraging the granularity of this dataset, we construct a firm-level measure of average cash flow expectations for professional forecasters based in Mainland China and Hong Kong. We then analyze the reaction of the ratio of these expectations to monetary policy surprises using the procedure described in Section 2.3. Appendix B provides details on the data construction and the results of this analysis. Consistent with our identification assumptions, we find that cash flow expectations are highly correlated across the two regions and observe no significant divergence in expectations following a monetary policy surprise.

5.2. Exchange rates

Another potential channel through which an unexpected shift in monetary policy could theoretically impact the A/H ratio is exchange rate fluctuations. As previously noted, H-shares are denominated in HKD, while A-shares are denominated in CNY. To ensure a consistent unit of measurement, we adjust the A/H ratio using the daily spot HKD-CNY exchange rate. If this exchange rate were to react to US monetary policy surprises, our estimates might inadvertently capture some of these exchange rate adjustments. Although prior studies have

²⁷ For example, Jia et al. (2017) show that heterogeneity in analyst forecasts of firm-specific cash flow expectations across markets contributes to price dispersion.

²⁸ Because earlier and more recent US export share data are unavailable, only announcements between Jan. 3, 2001, and Dec. 14, 2022, in Table 2 are included in the sample.

Table A.1

Robustness interacting with US export share.

	Full sample					Share = 0	Share > 0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Surprise $\times$ Post	0.213*** (0.0319)	0.199*** (0.0316)	0.200*** (0.0316)	0.201*** (0.0316)	0.196*** (0.0335)	0.194*** (0.0411)	0.208*** (0.0396)
US export share	-0.00644 (0.00835)	-0.00675 (0.00829)	0.0178 (0.0114)	0.0148** (0.00627)	0.0116** (0.00527)		
Surprise $\times$ US export share	-0.0662 (0.0466)	-0.0671 (0.0467)	-0.0199 (0.0339)	-0.0581** (0.0255)	-0.0225 (0.0211)		
Post $\times$ US export share	-0.0000270 (0.000254)	0.000537 (0.000372)	0.000531 (0.000370)	0.000607 (0.000380)	0.000606 (0.000388)		
Surprise $\times$ Post $\times$ US export share	-0.00155 (0.00601)	0.000126 (0.00607)	0.000174 (0.00608)	0.000319 (0.00609)	-0.0000369 (0.00604)		
Company FE	No	No	Yes	Yes	Yes	No	No
Event time	No	Post	Post	FE	FE	Post	Post
Company Spec. Trend	No	No	No	Common	Separated	No	No
Announcements	110	110	110	110	110	110	92
Observations	53,988	53,988	53,988	53,988	53,988	29,135	24,853

This table presents additional robustness tests, estimating Eq. (7) and variants of Eq. (3) split by *US Export Share*. Because earlier US export share data are not available, only announcements after January 3, 2001, in Table 2 are included in the sample in Columns (1)–(5), and only announcements after January 29, 2003, in Table 2 are included in the sample in Column (7). Columns (1)–(5) summarize the results of estimating Eq. (7). Column (6) summarizes the result of estimating Eq. (3) using the subsample of firms with zero US export share. Column (7) summarizes the result of estimating Eq. (3) using the subsample of firms with positive US export share. In each specification, we control for announcement fixed effects. *Company FE* represents whether we control for company fixed effects. *Event Time* indicates how event time effects are controlled for: *No* indicates that event time is not controlled for; *Post* indicates that a dummy variable for post-announcement event time is included; *FE* indicates that event time fixed effects are included. *Company Spec. Trend* represents whether and how company-specific time trends are controlled for: *No* indicates company-specific time trends are not included; *Common* indicates that we control for company-specific time trends common to surprise increases and decreases; *Separated* indicates that we control for company-specific time trends separated for surprise increases and decreases. *Announcements* represents the number of FOMC announcements. *Observations* represents the number of observations. Standard errors clustered at the company level are in parentheses. *** represents significance at the 1% level, ** 5% level, * 10% level.

documented exchange rate responses to monetary policy surprises, this is unlikely to be a significant concern in our context. Both Hong Kong and Mainland China have maintained a de facto peg to the USD, with Mainland China relying on foreign exchange interventions and capital controls as primary stabilization mechanisms while Hong Kong relies on interest rate adjustments. In Appendix C, we formally assess this issue through event studies of exchange rate movements following policy announcements. Figs. C.1, C.2, and C.3 show no significant changes in exchange rates or forward rates around these announcements, consistent with the presence of a stabilization regime.

5.3. Alternative specifications

Our main results remain robust across a range of alternative specifications. Table 7 presents findings from variations in the sample selection. Column (1) includes monetary policy announcements made at the zero lower bound, which are excluded from our main analysis. Columns (2), (3), and (4) sequentially exclude very small surprise announcements, announcements made during holidays, and those occurring on days surrounding holidays. Across all specifications, the results remain consistent with those reported in Table 3.

Another potential concern is that dynamic and heterogeneous treatment effects might bias our results. In Appendix D, we address this issue using a stacked estimator that is robust to such effects, following the approach summarized by Baker et al. (2022) and applied by Cengiz et al. (2019).

Additional robustness tests are presented in Online Appendix E. We demonstrate that our main findings are robust to alternative clustering methods (Tables E.8 and E.9), wild bootstrap techniques (Table E.10), fixed effects specifications (Table E.11), a wider event-time window (Table E.12), and sample restrictions (Tables E.13 and E.14). The results and their magnitudes remain consistent across these specifications. Additionally, Table E.15 provides a sensitivity analysis for potential outliers by reporting our main estimates after systematically excluding each individual announcement.

6. Concluding remarks

This paper uses monetary policy surprises and claims to the same cash flows to study how monetary policy impacts asset prices through discount factors. We provide conclusive evidence that monetary policy announcements cause investors to revise their discount factors and impact stock prices. Consistent with theories of time-varying perceptions about the level of activism from the Fed, this result is concentrated in surprises of a direction that amplifies the current cycle. Moreover, we find that the effect persists after the initial announcement, which suggests that higher-frequency strategies may underestimate the effects of policy transmission. While this paper highlights the critical role of the discount rate channel in the reaction of asset prices to monetary policy, future research could further explore the underlying drivers of the pronounced asymmetry between cycle-amplifying and cycle-contradicting surprises.

CRedit authorship contribution statement

Quentin Vandeweyer: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Minghao Yang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Constantine Yannelis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors have nothing to disclose.

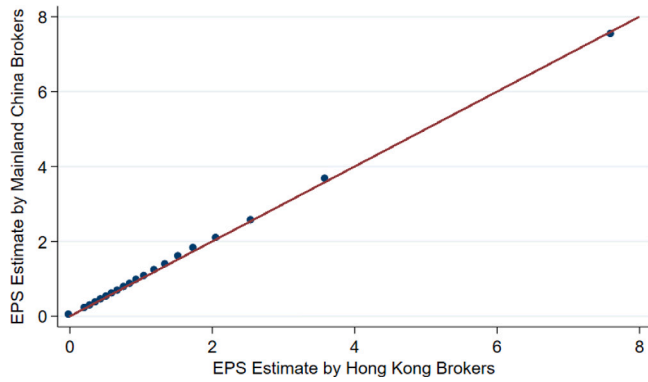


Fig. B.1. Binscatter plot of EPS forecasts.

This figure presents a binscatter plot of EPS forecasts by Mainland China brokers versus those by Hong Kong brokers. The blue dots are binned scatters of EPS forecasts, and the red line is the 45 degree line. The sample period spans from July 1993 to August 2024. Each underlying observation is defined as a stock-day for which there is no missing EPS forecast from either Hong Kong brokers or Mainland China brokers for the stock on that day, and there is an update in the forecast for the stock from at least one Hong Kong broker or mainland broker on that day. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

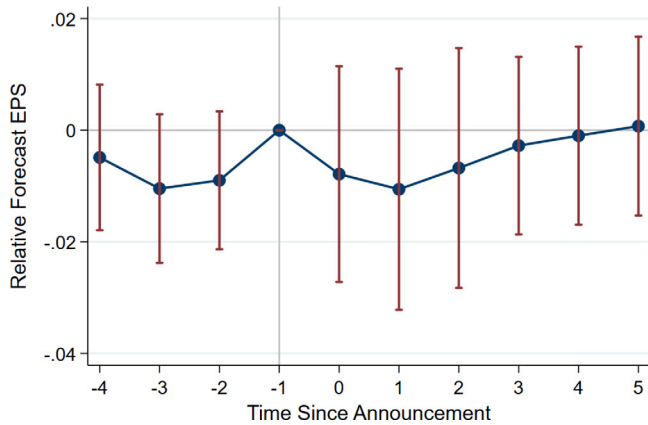


Fig. B.2. Policy announcements and EPS forecasts.

This figure plots estimates of a variant of Eq. (5) around FOMC announcements, where the dependent variable is the EPS forecast ratio of Mainland China brokers over Hong Kong brokers. Because EPS forecasts are infrequent, we drop announcements in Table 2 that are not associated with forecasts for any dual-listed stocks from both Hong Kong and Mainland China brokers. Company, announcement, and forecasting period fixed effects, as well as a post-announcement indicator, are included. Standard errors are clustered at the company level. The horizontal axis is the event-time indicator. The time since the announcement equals zero if the FOMC announcement was published on that date. Blue dots represent the regression coefficients, and red bands represent 95% confidence intervals. The vertical gray line marks time since announcement = -1, where we normalize the coefficients to zero. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Appendix A. Heterogeneity with US export shares

One potential concern in our analysis is that imperfect information flows or heterogeneous beliefs between Mainland China and Hong Kong could be driving part of our results. To assess these potential concerns,

we collect data on the shares of sales from export for the companies in our sample using data from FactSet GeoRev.²⁹ We perform the following triple-difference regression:

$$\begin{aligned} (P_A/P_H)_{ist} = & \alpha_i + \eta_s + \lambda_t + \beta_1 \text{Surprise}_s \times \text{Post}_t \\ & + \beta_2 \text{US Export Share}_{ist} + \beta_3 \text{Surprise}_s \times \text{US Export Share}_{ist} \\ & + \beta_4 \text{Post}_t \times \text{US Export Share}_{ist} + \beta_5 \text{Surprise}_s \times \text{Post}_t \\ & \times \text{US Export Share}_{ist} + \varepsilon_{ist} \end{aligned} \quad (7)$$

where $\text{US Export Share}_{ist}$ is firm i 's revenue share from the US on announcement s and at event time t . Other variables are the same as those defined in Eq. (3).

Besides the triple-difference specification, we also perform a difference-in-differences specification, splitting the sample by whether the company has a positive US export share. This allows us to examine whether the cash flow exposure to US monetary policy is driving the response of the A/H ratio to monetary surprises by comparing the responses of companies with zero and positive US export shares.

Table A.1 presents the results. Columns (1)–(5) present the triple-difference results with different fixed effects and trends included. The main interaction remains significant following the announcement, and we do not see significant differences in the interactions of export shares. Columns (6) and (7) present variants of the main results, restricting to a subsample with US export shares equal to zero and a subsample with US export shares greater than zero. The results are highly similar and statistically indistinguishable.

Appendix B. Monetary policy surprises and EPS forecasts

To further test whether heterogeneous beliefs about firms' cash flows following a monetary policy surprise are potentially driving some of our results, we use the earning-per-share (EPS) forecasts from professional forecasters on dual-listed stocks by both Hong Kong and Mainland China brokers and investigates how these forecasts differ following a monetary policy surprise.

Our EPS forecast data come from the I/B/E/S database. It provides recommendation scores and EPS forecasts on stocks traded in major stock exchanges all over the world, provided by analysts from various brokerage houses (hereafter brokers). I/B/E/S provides the last name, first initial, and ID for each analyst, the CUSIP of the target stock, the analyst's EPS forecast and the forecasting period, and a numeric ID for the broker the analyst works for. Observations in the EPS forecast dataset differ in three ways. We extract recommendation scores and EPS forecasts from 1993 to 2023 for the dual-listed stocks,³⁰ and prepare the data using the following steps.

The first step is to translate the numeric broker ID into the broker's name. We merge the EPS forecast dataset with the recommendation score dataset based on the analyst's ID and the recommendation date and time. This allows us to obtain the alphabetic broker ID. Unfortunately, I/B/E/S has stopped providing the translation file between the alphabetic ID and the broker's name.³¹ Therefore, we first employ Table A1.1 in Shen (2022), Table 2 in Hashim (2012), and Online

²⁹ For many companies in our sample, US export share data for earlier years are not available in FactSet. Therefore, we drop these observations in the following analysis. Because of data limitations, we only have the export shares data until 2022, so we restrict our sample to the announcements listed in Table 2 before 2023.

³⁰ The H- and A-share of the same stock have different CUSIPs, which allows us to distinguish different EPS forecasts between shares. Nevertheless, the forecast data suggest that if an analyst has made two EPS forecasts on the H- and A-share of the same stock at the same time, the forecasts will not differ in most cases. Therefore, we do not distinguish between shares in this exercise.

³¹ See Wharton Research Data Services (2025).

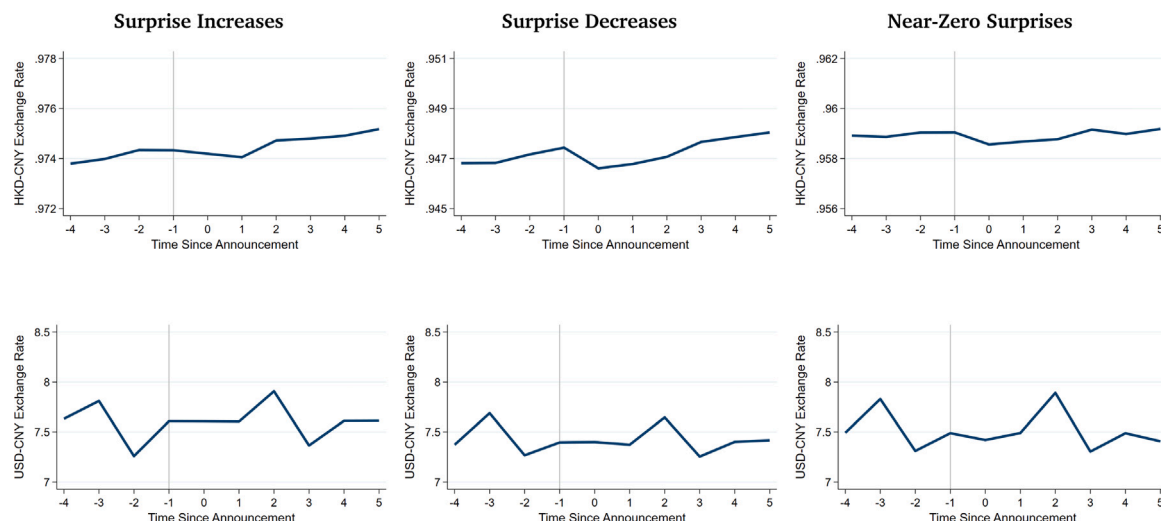


Fig. C.1. Means of exchange rates around surprise increases and decreases.

This figure plots the means of HKD-CNY and USD-CNY exchange rates around FOMC announcements, grouped by surprise increases ($Surprise \geq 0.01$), surprise decreases ($Surprise \leq -0.01$), and near-zero surprises ($-0.01 < Surprise < 0.01$). All announcements listed in Table 2 are included in the sample. An announcement is considered a surprise increase if Surprise is larger than 0.01. An announcement is considered a surprise decrease if Surprise is smaller than -0.01 . An announcement is considered a near-zero surprise if the Surprise is larger than -0.01 and smaller than 0.01. The left column plots the means of the exchange rates around surprise increases; the middle column plots the means around surprise decreases; and the right column plots the means around near-zero surprises. In each column, the top panel plots the mean of the HKD-CNY exchange rate, and the bottom panel plots the mean of the USD-CNY exchange rate. The horizontal axis is the event-time indicator. The time since the announcement equals zero if the FOMC announcement was published on that date. The vertical gray line marks time since announcement = -1 .

Appendix Table A1.1 in Law (2023), in which the translation of several brokers is provided, to identify most of the brokers in our sample. For the remaining brokers, Section A1.3 in Shen (2022) points out that some alphabetic IDs are abbreviations of the broker's full name (for instance, GOLDMAN stands for Goldman Sachs), so we manually matched each couple of alphabetic IDs and analyst name to the most plausible brokerage firm. Following the above procedure, we are able to identify 151 of 196 brokers, which corresponds to more than 97% of the forecasts in our sample.

The second step is to determine whether the broker is located in Hong Kong, Mainland China, or elsewhere. We manually collect each broker's registration country or region from Bloomberg and only keep the forecasts by Hong Kong brokers or Mainland China brokers. This leaves us with 34 in Hong Kong and 47 in Mainland China.

The third step is to clean the EPS forecasts. First, forecasts vary in their forecasting periods. We observe that more than 95% of the forecasts are for the current fiscal year, the next fiscal year, or the second subsequent fiscal year. Therefore, we focus on those forecasting horizons. Second, the forecasts can be either primary EPS or diluted EPS. To keep things comparable, we focus on the primary EPS forecasts for all stocks, which account for around 85% of the sample. Third, the forecasting period indicator and the forecasting period end date are sometimes inconsistent. We drop those observations from our sample. Fourth, the forecasts are made in either HKD or CNY. We use the HKD-CNY daily exchange rate to convert HKD forecasts into CNY.

The fourth step is to aggregate the data to the region-by-stock-by-day level. This aggregation is necessary because most brokers in our sample do not frequently make EPS forecasts. Without appropriate aggregation, we would not observe much variability in the EPS forecasts after we merge the EPS forecast data with our baseline event study data. We first aggregate the analyst forecasts to the broker-by-stock-by-day level. Specifically, when any analyst from a broker makes a new forecast on a given day for a given stock, we treat that forecast as the broker's forecast for that stock from that day until a new forecast for the stock is made by some analyst from the same broker. If multiple analysts from a broker are making forecasts on the same day for the

same stock, we use the average forecast as the broker's forecast. Next, we aggregate broker forecasts to the region by stock by day level. For a given day, we calculate the average forecast with the same forecasting period of all Hong Kong (Mainland China) brokers on every stock and use the average as the forecast of Hong Kong (Mainland China) brokers on the corresponding stock and for the corresponding forecasting period.

Fig. B.1 depicts a binscatter plot of EPS forecasts by Mainland China brokers versus those by Hong Kong brokers using the aggregated forecast data. Each observation is a stock-day for which there is no missing EPS forecast from either Hong Kong brokers or Mainland China brokers and for which there is a forecast update from at least one broker on that day. The figure suggests that the forecasts are highly correlated across regions.

Next, we merge the aggregated forecast data with the event study data and conduct an event study similar to our baseline specification for the A/H stock price ratio. We change the dependent variable in Eqs. (3) and (5) to the ratio of EPS forecasts by Mainland China brokers over those by Hong Kong brokers. We winsorize the forecasts at the 5% level on both sides to alleviate the impact of outliers. To account for the difference in the forecasting period, we add forecasting period fixed effects to the equations. Fig. B.2 plots the event study coefficients from Eq. (5), and Table B.1 summarizes the difference-in-differences results from Eq. (3). Both of these suggest that the relative EPS forecasts do not change a lot following a monetary policy surprise, which suggests that beliefs about cash flow expectations across Hong Kong and Mainland China investors do not diverge much.

Appendix C. Exchange rates around FOMC announcement dates

In our baseline analysis, we quote H-share prices in CNY. However, in response to a surprise change in the target rate, it is possible that the exchange rate between HKD and CNY also changes. In this section, we conduct event studies on exchange rates to tackle this issue. We primarily focus on the exchange rate between HKD and CNY, but we also provide an analysis of that between USD and CNY since the

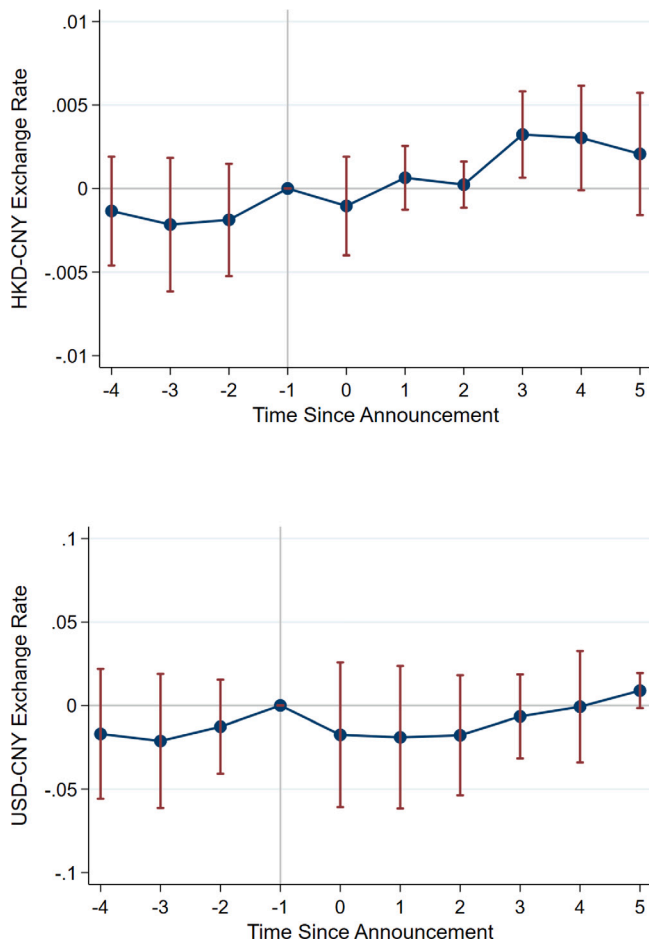


Fig. C.2. Policy announcements and exchange rates.

This figure plots estimates of Eq. (4) around FOMC announcements, where the dependent variable is either the HKD-CNY exchange rate or the USD-CNY exchange rate. All announcements listed in Table 2 are included in the sample. The vertical axis depicts either the HKD-CNY exchange rate or the USD-CNY exchange rate. Announcement fixed effects and a post-announcement indicator are included. Standard errors are clustered at the announcement level. In the top panel, the dependent variable is the HKD-CNY exchange rate. In the bottom panel, the dependent variable is the USD-CNY exchange rate. The horizontal axis is the event-time indicator. The time since the announcement equals zero if the FOMC announcement was published on that date. Blue dots represent the regression coefficients, and red bands represent 95% confidence intervals. The vertical gray line marks time since announcement = -1, where we normalize the coefficients to zero. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

exchange rate between USD and HKD is fixed. We measure exchange rates as the amount of CNY that one unit of HKD/USD can exchange, so an increase in exchange rates indicates an appreciation of HKD/USD. HKD-CNY exchange rate data are extracted from Wind, as in Section 3. USD-CNY exchange rate data are extracted from the FRED.

Fig. C.1 displays the means of HKD-CNY and USD-CNY exchange rates around announcements, grouped by surprise increases and decreases. The left column shows the raw means for surprise increases, the middle column for surprise decreases, and the right column for near-zero surprises. In each column, the upper graph shows HKD-CNY exchange rates, and the bottom panel shows USD-CNY exchange rates. Note that there are 23 increases, with an average surprise of 0.050, and 27 decreases, with an average surprise of -0.116. We see that around increases, the exchange rates change only marginally. However, around decreases, both HKD and USD depreciate. The following event studies

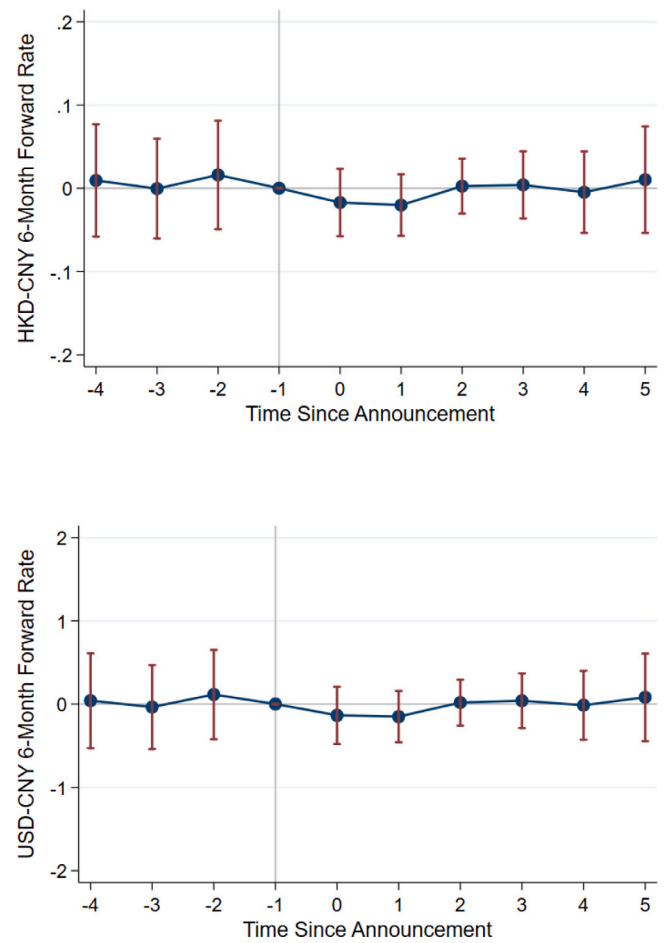


Fig. C.3. Policy Announcements and forward exchange rates.

This figure plots estimates of Eq. (4) around FOMC announcements, where the dependent variable is either the HKD-CNY 6-month-forward rate or USD-CNY 6-month-forward rate. Because earlier forward exchange rates are not available, only announcements after December 16, 2015, are included in the sample. The vertical axis depicts either the HKD-CNY 6-month-forward rate or the USD-CNY 6-month-forward rate. Announcement fixed effects and a post-announcement indicator are included. Standard errors are clustered at the announcement level. In the top panel, the dependent variable is the HKD-CNY 6-month-forward rate. In the bottom panel, the dependent variable is the USD-CNY 6-month-forward rate. The horizontal axis is the event-time indicator. The time since the announcement equals zero if the FOMC announcement was published on that date. Blue dots represent the regression coefficients, and red bands represent 95% confidence intervals. The vertical gray line marks time since announcement = -1, where we normalize the coefficients to zero. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

investigate whether there is a significant relationship between surprises and exchange rate changes.

Fig. C.2 displays the event studies on exchange rates with respect to surprises following the specification in the main text. The top panel plots the event study for the HKD-CNY exchange rate, and the bottom panel plots the USD-CNY exchange rate. We find that both exchange rates do not significantly change in response to surprises. This alleviates potential concerns regarding converting currency units when calculating the A/H ratio.

Another potential concern is that even though the exchange rates do not respond to monetary policy surprises immediately, they respond later, which affects the calculation of A/H ratios in the future. To

Table B.1

Monetary policy surprises and target rate and EPS forecasts.

	(1)	(2)	(3)	(4)	(5)
Surprise $\times$ Post	-0.00109 (0.00584)	0.00148 (0.00589)	0.00179 (0.00589)	0.00223 (0.00592)	0.00488 (0.00539)
Company FE	No	No	Yes	Yes	Yes
Event time	No	Post	Post	FE	FE
Company Spec. Trend	No	No	No	Common	Separated
Announcements	113	113	113	113	113
Observations	195,948	195,948	195,948	195,948	195,948

This table presents the results of estimating variants of Eq. (3). The dependent variable is the EPS forecast ratio of Mainland China brokers over Hong Kong brokers. Because EPS forecasts are infrequent, we drop announcements in Table 2 that are not associated with forecasts for any dual-listed stocks from both Hong Kong and Mainland China brokers. In each specification, we control for announcement and forecasting period fixed effects. *Company FE* represents whether we control for company fixed effects. *Event Time* indicates how event time effects are controlled for: *No* indicates that event time is not controlled for; *Post* indicates that a dummy variable for post-announcement event time is included; *FE* indicates that event time fixed effects are included. *Company Spec. Trend* represents whether and how company-specific time trends are controlled for: *No* indicates company-specific time trends are not included; *Common* indicates that we control for company-specific time trends common to surprise increases and decreases; *Separated* indicates that we control for company-specific time trends separated for surprise increases and decreases. *Announcements* represents the number of FOMC announcements. *Observations* represents the number of observations. Standard errors clustered at the company level are in parentheses. *** represents significance at the 1% level, ** 5% level, * 10% level.

alleviate this concern, we collect data on HKD-CNY and USD-CNY 6-month-forward exchange rates and conduct an analysis to examine their responses to monetary policy surprises. Forward exchange rate data are extracted from China Foreign Exchange Trade System (2024), and the time series starts from 2015.

Fig. C.3 displays the event studies on forward exchange rates with respect to surprises following Eq. (4). The top panel plots the event study for the HKD-CNY 6-month-forward rate and the bottom panel plots that for the USD-CNY 6-month-forward rate. The results imply that both forward exchange rates do not significantly change in response to surprises. This illustrates that the exchange rates are not changing in response to monetary policy surprises in the future, and thus alleviates concern that the A/H ratio might be miscalculated due to converting currency units.

Appendix D. Dynamic and heterogeneous treatment effects

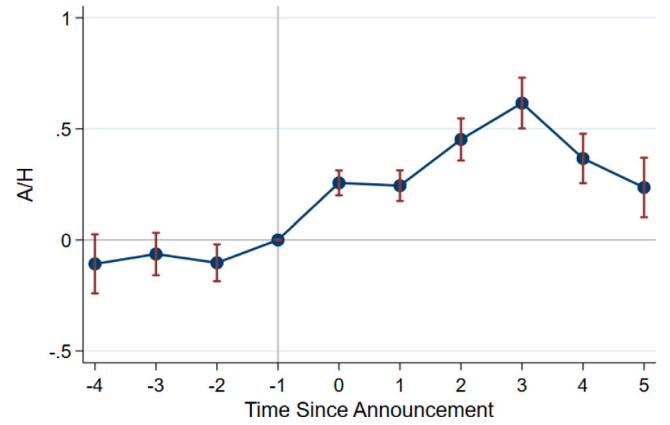
A possible concern is that dynamic and heterogeneous treatment effects could bias our results. To investigate whether our baseline results are driven by dynamic and heterogeneous treatment effects, we follow the stacked approach summarized by Baker et al. (2022) and applied by Cengiz et al. (2019).

To apply the stacked method to our setting, we first set the event time window to 5 periods before and after the announcement ($t \in \{-5, -4, \dots, 5\}$). We use the baseline 129 announcements from June 2000 to September 2024, when the zero lower bound was not reached, as the event set. Note that every stock in the sample during the event time window was affected by monetary policy surprises, so we used a different method to construct a control group. For each stock i at announcement s and event time t , we still used stock i as its control group but lagged it by 10 trading days from the trading day prior to announcement s and event time t . Note that this stacked approach almost doubles our previous sample size; however, several stocks do not have earlier price data because of initial issuance. We drop these observations.

We estimate the following specification:

$$(P_A/P_H)_{i(T),s,t} = \alpha_{i(T),s} + \eta_{s,t} + \beta \text{Surprise}_{i(T),s} \times \text{Post}_t + \varepsilon_{i(T),s,t} \quad (8)$$

where i stands for a stock, $T = 1$ if stock i is not lagged (treatment group) and $T = 0$ if it is lagged (control group), s stands for an announcement, and t stands for an event time. $\alpha_{i(T),s}$ is the announcement

**Fig. D.1.** Stacked policy announcements and share prices.

This figure plots estimates of Eq. (9) around FOMC announcements, where the dependent variable is the A/H share price ratio. All announcements listed in Table 2 are included in the sample. Company by treatment group by announcement and announcement by event time fixed effects are included. Standard errors are clustered at the company level. The horizontal axis is the event-time indicator. The time since the announcement equals zero if the FOMC announcement was published on that date. Blue dots represent the regression coefficients, and red bands represent 95% confidence intervals. The vertical gray line marks time since announcement = -1, where we normalize the coefficients to zero. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table D.1

Stacked main results.

	(1)	(2)	(3)	(4)
Surprise $\times$ Post	0.446*** (0.0405)	0.445*** (0.0404)	0.447*** (0.0408)	0.445*** (0.0409)
Company FE	No	Yes	Yes	Yes
Company Spec. Trend	No	No	Common	Separated
Announcements	129	129	129	129
Observations	186,232	186,232	186,232	186,232

This table presents the results of estimating Eq. (8). All announcements listed in Table 2 are included in the sample. In each specification, we control for announcement by treatment group and announcement by event time fixed effects. *Company FE* represents whether we control for company fixed effects: *No* implies that announcement by treatment group fixed effects are not interacted with company fixed effects; *Yes* implies that announcement by treatment group fixed effects are interacted with company fixed effects. *Company Spec. Trend* represents whether and how company-specific time trends are controlled for; *No* indicates company-specific time trends are not included; *Common* indicates that we control for company-specific time trends common to surprise increases and decreases; *Separated* indicates that we control for company-specific time trends separated for surprise increases and decreases. *Announcements* represents the number of FOMC announcements. *Observations* represents the number of observations. Standard errors clustered at the company level are in parentheses. *** represents significance at the 1% level, ** 5% level, * 10% level.

by stock group fixed effect, and $\eta_{s,t}$ is the announcement by event time fixed effect. The term $\text{Surprise}_{i(T),s}$ equals the Kuttner surprise of announcement s if $T = 1$ and zero otherwise, and Post_t equals one if t is larger than or equal to 0 and zero otherwise.

The event study specification is analogous and given by

$$(P_A/P_H)_{i(T),s,t} = \alpha_{i(T),s} + \eta_{s,t} + \sum_{\tau=-4}^5 \beta_{\tau} \text{Surprise}_{i(T),s} \times \mathbb{1}(\text{Time Since Announcement}_{st} = \tau) + \varepsilon_{i(T),s,t} \quad (9)$$

where $\mathbb{1}(\text{Time Since Announcement}_{st} = \tau)$ equals one if time since announcement equals τ and zero otherwise. The remaining variables are the same as those in Eq. (8).

Table D.1 presents difference-in-differences results following Eq. (8) and Fig. D.1 presents event study results following Eq. (9). The effects

are broadly similar to our main estimates, albeit with slightly larger coefficients.

Appendix E. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfineco.2025.104190>.

Data availability

Replication Package for Discount Factors and Monetary Policy: Evidence from Dual-Listed Stocks (pseudo-data) (Reference data) (Mendeley Data)

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